

Matrix-valued Factor Model with Time-varying Main Effects

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Abstract

We introduce the matrix-valued time-varying Main Effects Factor Model (MEFM). MEFM is a generalization to the traditional matrix-valued factor model (FM). We give rigorous definitions of MEFM and its identifications, and propose estimators for the time-varying grand mean, row and column main effects, and the row and column factor loading matrices for the common component. Rates of convergence for different estimators are spelt out, with asymptotic normality shown. The core rank estimator for the common component is also proposed, with consistency of the estimators presented. We propose a test for testing if FM is sufficient against the alternative that MEFM is necessary, and demonstrate the power of such a test in various simulation settings. We also demonstrate numerically the accuracy of our estimators in extended simulation experiments. A set of NYC Taxi traffic data is analysed and our test suggests that MEFM is indeed necessary for analysing the data against a traditional FM.

Key words and phrases: Large-scale dependent data; Time-varying row and column effects; MEFM and FM interchange; sufficiency of FM over MEFM; Tucker decomposition.

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1 Introduction

Matrix-valued time series factor models, a generalization of vector time series factor models (Bai, 2003; Stock and Watson, 2002), have been utilized a lot for dimension reduction and prediction in recent years in fields such as finance, economics, medical science and meteorology, to name but a few. This is a subject still in its infancy, but important earlier theoretical and methodological developments include Wang et al. (2019), Chen et al. (2020), Chen and Fan (2023) and He et al. (2024), which are all on matrix-valued factor models using the Tucker decomposition for the common component. Chang et al. (2023) uses the CP decomposition for the common component, while Guan (2023) considers Tucker decomposition of the common component but taking in covariates in the loadings. Beyond factor modelling, Chen et al. (2021), Wu and Bi (2023) and Zhang (2024) propose autoregressive and moving average models for matrix-valued time series data. See Tsay (2023) for a comprehensive review of matrix-valued time series analysis.

With Tucker decomposition, a matrix-valued time series factor model (FM) can be written as

$$\mathbf{Y}_t = \boldsymbol{\mu} + \mathbf{R}\mathbf{F}_t\mathbf{C}' + \mathbf{E}_t, \quad (1.1)$$

where $\mathbf{Y}_t \in \mathbb{R}^{p \times q}$ is the observed matrix at time t , $\boldsymbol{\mu} \in \mathbb{R}^{p \times q}$ is the mean matrix, $\mathbf{R} \in \mathbb{R}^{p \times k_r}$ and $\mathbf{C} \in \mathbb{R}^{q \times k_c}$ are the row and column factor loading matrices respectively, $\mathbf{F}_t \in \mathbb{R}^{k_r \times k_c}$ is the core factor matrix at time t , and finally $\mathbf{E}_t \in \mathbb{R}^{p \times q}$ is the noise matrix at time t . If we set $\mathbf{R} := (\boldsymbol{\alpha}_{p \times r}, \widetilde{\mathbf{R}}_{p \times (k_r - r - \ell)}, \mathbf{1}_{p \times \ell})$, $\mathbf{C} := (\mathbf{1}_{q \times r}, \widetilde{\mathbf{C}}_{q \times (k_c - r - \ell)}, \boldsymbol{\gamma}_{q \times \ell})$ and $\mathbf{F}_t := \text{diag}((\mathbf{g}_t)_{r \times r}, (\widetilde{\mathbf{F}}_t)_{(k_r - r - \ell) \times (k_c - r - \ell)}, (\mathbf{h}_t)_{\ell \times \ell})$ (He et al., 2023), where $\mathbf{1}_{m \times n}$ is a matrix of ones of size $m \times n$, then (1.1) becomes

$$\mathbf{Y}_t = \boldsymbol{\mu} + \boldsymbol{\alpha}\mathbf{g}_t\mathbf{1}_{r \times q} + \mathbf{1}_{p \times \ell}\mathbf{h}_t\boldsymbol{\gamma}' + \widetilde{\mathbf{R}}\widetilde{\mathbf{F}}_t\widetilde{\mathbf{C}}' + \mathbf{E}_t. \quad (1.2)$$

If the rows of $\mathbf{Y}_t$ represent different countries and the columns represent different economic indicators, then since the j th row of $\boldsymbol{\alpha}\mathbf{g}_t\mathbf{1}_{r \times q}$ is $\boldsymbol{\alpha}_j \cdot \mathbf{g}_t\mathbf{1}_{r \times q}$, where $\boldsymbol{\alpha}_j \cdot$ is the j th row of $\boldsymbol{\alpha}$, it means that each element in the j th row is the same, with value $\boldsymbol{\alpha}_j \cdot \mathbf{g}_t\mathbf{1}_r$. Hence we can argue that $\mathbf{g}_t$ represents common global factors affecting all countries, although each country loads differently on $\mathbf{g}_t$. Similarly, $\mathbf{h}_t$ represents latent economic states across different economic indicators. The term $\widetilde{\mathbf{R}}\widetilde{\mathbf{F}}_t\widetilde{\mathbf{C}}'$ can be viewed as an interaction term., while $\boldsymbol{\alpha}\mathbf{g}_t\mathbf{1}_{r \times q}$ the country main effects, and $\mathbf{1}_{p \times \ell}\mathbf{h}_t\boldsymbol{\gamma}'$ the economic states' main effects.

Three problems arise upon inspecting (1.1) and (1.2) however. Firstly, for (1.1) to transform to (1.2), $\mathbf{R}$ and $\mathbf{C}$ are both of reduced rank. In the literature for model (1.1), we always need $\mathbf{R}$ and $\mathbf{C}$ to be of full rank at least asymptotically (see for example, Assumption (B2) in He et al. (2023) or Equation (8) in Chen and Fan (2023)) for estimation purpose.

Secondly, model (1.2) is not general enough, unless r and ℓ can be large. For example, if r is small,

each country is driven only by few global common factors affecting all countries, on top of the factors in $\tilde{\mathbf{F}}_t$. This will not be a problem, if not for the fact that there can be latent common factors that only drive a small group of countries/economic indicators. For instance, there can be a few small European countries which do not share global common factors with the majority of European countries, but with other middle-Eastern countries. Such “grouping” of countries usually comes with their corresponding groups of unique factors. These unique factors become “weak” country effects, shared only among “small” number of countries, essentially inflating the value of r while inducing a sparse $\boldsymbol{\alpha}$. Constraint factor modelling in Chen et al. (2020) can certainly help, but we do not always know the exact group of countries which share latent common factors.

The final problem is related to the second one. The inability of (1.2) to accommodate “weak” country/economic states effects originates from the fact that the common component in (1.1), $\tilde{\mathbf{R}}\tilde{\mathbf{F}}_t\tilde{\mathbf{C}}'$, contains only pervasive factors, which is essentially assumed across all past works in factor models for matrix-valued time series. In a tensor setting however, Cen and Lam (2024) and Chen and Lam (2024) have both allowed weak factors in the common component in the factor model.

One way to generalize (1.2) to address all aforementioned problems is to note that

$$\boldsymbol{\alpha}\mathbf{g}_t\mathbf{1}_{r\times q} = (\boldsymbol{\alpha}\mathbf{g}_t\mathbf{1}_r)\mathbf{1}_q' =: \boldsymbol{\alpha}_t\mathbf{1}_q', \quad \mathbf{1}_{p\times \ell}\mathbf{h}_t\boldsymbol{\gamma}' = \mathbf{1}_p(\boldsymbol{\gamma}\mathbf{h}_t'\mathbf{1}_\ell)' =: \mathbf{1}_p\boldsymbol{\beta}_t',$$

where $\boldsymbol{\alpha}_t$ and $\boldsymbol{\beta}_t$ are the time-varying row and column main effects respectively. If we are able to estimate the two vectors $\boldsymbol{\alpha}_t$ and $\boldsymbol{\beta}_t$ without any low-rank constraints as in the equation above, then the second problem is naturally solved. Formally allowing for weak factors in the row and column loading matrices, like those in Lam and Yao (2012) for a vector factor model, solves the third problem. Finally, with these problems solved, we can go back to assuming full rank row and column factor loading matrices (see Assumption (L1) in Section 4.1 in this paper) to solve the first problem.

In this paper, we contribute to the literature in several important ways. Firstly, we generalize model (1.2) to (3.1) which is the time-varying main effects factor model (MEFM), incorporating all relaxations described in the previous paragraph. Secondly, we provide estimation and inference methods and the corresponding theoretical guarantees, on top of a separate ratio-based method for identifying the core rank of $\mathbf{F}_t$, with consistency proved. Third and perhaps the most important of all, we provide a statistical test to test if FM in (1.1), with $\mathbf{R}$ and $\mathbf{C}$ both of full rank, is sufficient against the more general MEFM in (3.1). A rejected null hypothesis of FM being sufficient for the data then means that there are row and/or column main effects that is not of a low rank structure like those in (1.2), essentially pointing to the existence of “weak” main effects.

The rest of the paper is organised as follows. Section 2 introduces the notations used in this paper.

Section 3 introduces MEFM formally, laying down important identification conditions and estimation methodologies for all the components in the model. Section 4 presents the assumptions for MEFM and the consistency and asymptotic normality results for its estimators. The test for FM versus MEFM is detailed in Section 4.5, while the core rank estimator for $\mathbf{F}_t$ is presented in Section 4.6. Finally, Section 5 presents our extensive simulation results and details the NYC Taxi traffic data analysis, pinpointing the presence of weak hourly main effects in the data. Our method is available in the R package **MEFM**, with instruction in its reference manual on CRAN. All proofs of the theorems are relegated to the supplementary materials of this paper.

2 Notations

Throughout this paper, we use the lower-case letter, bold lower-case letter and bold capital letter, i.e. $a, \mathbf{a}, \mathbf{A}$, to denote a scalar, a vector and a matrix respectively. We also use $a_i, A_{ij}, \mathbf{A}_{i\cdot}, \mathbf{A}_{\cdot i}$ to denote, respectively, the i -th element of $\mathbf{a}$, the (i, j) -th element of $\mathbf{A}$, the i -th row vector (as a column vector) of $\mathbf{A}$, and the i -th column vector of $\mathbf{A}$. We use $\circ$ to denote the Hadamard product. We use $a \asymp b$ to denote $a = O(b)$ and $b = O(a)$, while $a \asymp_P b$ to denote $a = O_P(b)$ and $b = O_P(a)$.

Given a positive integer m , define $[m] := \{1, 2, \dots, m\}$. The vector $\mathbf{1}_m$ denotes a vector of ones of length m . The i -th largest eigenvalue of a matrix $\mathbf{A}$ is denoted by $\lambda_i(\mathbf{A})$. The notation $\mathbf{A} \succcurlyeq 0$ (resp. $\mathbf{A} \succ 0$) means that $\mathbf{A}$ is positive semi-definite (resp. positive definite). We use $\mathbf{A}'$ to denote the transpose of $\mathbf{A}$, and $\text{diag}(\mathbf{A})$ to denote a diagonal matrix with the diagonal elements of $\mathbf{A}$, while $\text{diag}(\{a_1, \dots, a_n\})$ represents the diagonal matrix with $\{a_1, \dots, a_n\}$ on the diagonal.

Some norm notations. For a given set, we denote by $|\cdot|$ its cardinality. We use $\|\cdot\|$ to denote the spectral norm of a matrix or the L_2 norm of a vector, and $\|\cdot\|_F$ to denote the Frobenius norm of a matrix. We use $\|\cdot\|_{\max}$ to denote the maximum absolute value of the elements in a vector or a matrix. The notations $\|\cdot\|_1$ and $\|\cdot\|_\infty$ denote the L_1 and L_∞ -norm of a matrix respectively, defined by $\|\mathbf{A}\|_1 := \max_j \sum_i |(\mathbf{A})_{ij}|$ and $\|\mathbf{A}\|_\infty := \max_i \sum_j |(\mathbf{A})_{ij}|$. WLOG, we always assume the eigenvalues of a matrix are arranged by descending orders, and so are their corresponding eigenvectors.

3 Model and Estimation

3.1 Main effect matrix factor model

We propose the time-varying **Main Effect matrix Factor Model** (MEFM) such that for $t \in [T]$,

$$\mathbf{Y}_t = \mu_t \mathbf{1}_p \mathbf{1}_q' + \boldsymbol{\alpha}_t \mathbf{1}_q' + \mathbf{1}_p \boldsymbol{\beta}_t' + \mathbf{C}_t + \mathbf{E}_t, \quad (3.1)$$

where $\mathbf{Y}_t$ is a $p \times q$ observed matrix at time t , μ_t is a scalar representing the grand mean of $\mathbf{Y}_t$, $\boldsymbol{\alpha}_t \in \mathbb{R}^p$ and $\boldsymbol{\beta}_t \in \mathbb{R}^q$ are the row and column main effects at time t , respectively. The common component $\mathbf{C}_t := \mathbf{A}_r \mathbf{F}_t \mathbf{A}_c'$ is latent, where $\mathbf{F}_t \in \mathbb{R}^{k_r \times k_c}$ is the core factor series with unknown number of factors k_r and k_c , and $\mathbf{A}_r$ and $\mathbf{A}_c$ are the row and column factor loading matrices, with size $p \times k_r$ and $q \times k_c$, respectively. Lastly, $\mathbf{E}_t$ is the idiosyncratic noise series with the same dimension as $\mathbf{Y}_t$.

Unlike FM in (1.2), the main effects α_t and β_t in MEFM are not restricted to be of low rank, which significantly improves the flexibility of FM, and allows for a test of FM in (1.1) in the end. In fact, setting concatenated matrices $\ddot{\mathbf{A}}_r = (\mathbf{1}_p, \mathbf{I}_p, \mathbf{A}_r, \mathbf{1}_p)$ and $\ddot{\mathbf{A}}_c = (\mathbf{1}_q, \mathbf{1}_q, \mathbf{A}_c, \mathbf{I}_q)$, block diagonal matrix $\ddot{\mathbf{F}}_t = \text{diag}\{\mu_t, \boldsymbol{\alpha}_t, \mathbf{F}_t, \boldsymbol{\beta}_t\}$, then we can read (3.1) as

$$\mathbf{Y}_t = \ddot{\mathbf{A}}_r \ddot{\mathbf{F}}_t \ddot{\mathbf{A}}_c' + \mathbf{E}_t.$$

However, we observe that the dimension of the factor series is now $(2+p+k_r) \times (2+q+k_c)$, and hence there is not much dimension reduction for $\mathbf{Y}_t$, and both $\ddot{\mathbf{A}}_r$ and $\ddot{\mathbf{A}}_c$ have no full column ranks. This observation suggests again that MEFM is more general than FM, and numerical results in Section 5 actually show that even an approximate estimation by FM in general comes at a cost of using very large number of factors.

Given the above motivation of MEFM, we point out that the form of MEFM can be obtained by FM in general, see Remark 1 for details. For generality purpose, $\mathbf{Y}_t$ can have non-zero mean but we can always demean the data as the sample mean is not our main parameter of interest. The right hand side of (3.1) is entirely latent and hence we propose Assumption (IC1) below to identify the grand mean and the row and column effects.

(IC1) (Identification) *For any $t \in [T]$, we assume $\mathbf{1}_p' \boldsymbol{\alpha}_t = \mathbf{1}_q' \boldsymbol{\beta}_t = 0$, $\mathbf{1}_p' \mathbf{A}_r = \mathbf{0}$ and $\mathbf{1}_q' \mathbf{A}_c = \mathbf{0}$.*

However, we require further identification between the factors and the factor loading matrices. To do this, we normalise the loading matrices to $\mathbf{Q}_r = \mathbf{A}_r \mathbf{Z}_r^{-1/2}$ and $\mathbf{Q}_c = \mathbf{A}_c \mathbf{Z}_c^{-1/2}$, where $\mathbf{Z}_r = \text{diag}(\mathbf{A}_r' \mathbf{A}_r)$ and $\mathbf{Z}_c = \text{diag}(\mathbf{A}_c' \mathbf{A}_c)$, measuring the sparsity of each column of loading matrices and hence the factor strength. For example, $\mathbf{F}_t$ pervasive in the j -th row will have the j -th column of $\mathbf{A}_r$ dense and hence the j -th diagonal entry of $\mathbf{Z}_r$ will be of order p . For technical details, see Assumption (L1). We leave the identification to Section 4.1. Assumption (IC1) also facilitates the estimation of μ_t , $\boldsymbol{\alpha}_t$ and $\boldsymbol{\beta}_t$, and we discuss in the next section how to estimate the grand mean, the row and column effects, and the row and column factor loading matrices in (3.1).

3.2 Estimation of the main effects and factor components

The factor structure is hidden in $\mathbf{Y}_t$ and we need to estimate the time-varying grand mean and main effects first. For the grand mean, right-multiplying by $\mathbf{1}_q$ and left-multiplying by $\mathbf{1}_p'$ on both sides of (3.1) results in $\mathbf{1}_p' \mathbf{Y}_t \mathbf{1}_q = pq\mu_t + \mathbf{1}_p' \mathbf{E}_t \mathbf{1}_q$ by Assumption (IC1). Hence for each $t \in [T]$, we obtain the moment estimator for the time-varying grand mean as

$$\hat{\mu}_t := \mathbf{1}_p' \mathbf{Y}_t \mathbf{1}_q / pq. \quad (3.2)$$

Also, right-multiplying by $\mathbf{1}_q$ and left-multiplying by $\mathbf{1}_p'$ lead respectively to $\mathbf{Y}_t \mathbf{1}_q = q\mu_t \mathbf{1}_p + q\boldsymbol{\alpha}_t + \mathbf{E}_t \mathbf{1}_q$ and $\mathbf{1}_p' \mathbf{Y}_t = p\mu_t \mathbf{1}_q' + p\boldsymbol{\beta}_t' + \mathbf{1}_p' \mathbf{E}_t$. Therefore, we obtain the time-varying row and column effect estimators as

$$\hat{\boldsymbol{\alpha}}_t := q^{-1} \mathbf{Y}_t \mathbf{1}_q - \hat{\mu}_t \mathbf{1}_p, \quad \hat{\boldsymbol{\beta}}_t' := p^{-1} \mathbf{1}_p' \mathbf{Y}_t - \hat{\mu}_t \mathbf{1}_q'. \quad (3.3)$$

Finally, we introduce the following to estimate the factor structure,

$$\begin{aligned} \hat{\mathbf{L}}_t &:= \mathbf{Y}_t - \hat{\mu}_t \mathbf{1}_p \mathbf{1}_q' - \hat{\boldsymbol{\alpha}}_t \mathbf{1}_q' - \mathbf{1}_p \hat{\boldsymbol{\beta}}_t' = \mathbf{Y}_t + (pq)^{-1} \mathbf{1}_p' \mathbf{Y}_t \mathbf{1}_q \mathbf{1}_p \mathbf{1}_q' - q^{-1} \mathbf{Y}_t \mathbf{1}_q \mathbf{1}_q' - p^{-1} \mathbf{1}_p \mathbf{1}_p' \mathbf{Y}_t \\ &= \mathbf{M}_p \mathbf{Y}_t \mathbf{M}_q, \end{aligned} \quad (3.4)$$

where $\mathbf{M}_m := \mathbf{I}_m - m^{-1} \mathbf{1}_m \mathbf{1}_m'$ for any positive integer m . From the above, $\hat{\mathbf{L}}_t \hat{\mathbf{L}}_t'$ admits $\mathbf{1}_p$ in its null space, and $\hat{\mathbf{L}}_t' \hat{\mathbf{L}}_t$ admits $\mathbf{1}_q$ instead. The factor structure can hence be estimated, with $\hat{\mathbf{Q}}_r$ constructed as the eigenvectors of $T^{-1} \sum_{t=1}^T \hat{\mathbf{L}}_t \hat{\mathbf{L}}_t'$ corresponding to the first k_r largest eigenvalues, and $\hat{\mathbf{Q}}_c$ the eigenvectors of $T^{-1} \sum_{t=1}^T \hat{\mathbf{L}}_t' \hat{\mathbf{L}}_t$ corresponding to the first k_c largest eigenvalues.

We can then estimate the factor time series $\mathbf{F}_{Z,t} = \mathbf{Z}_r^{1/2} \mathbf{F}_t \mathbf{Z}_c^{1/2}$, and the common component $\mathbf{C}_t$, respectively as

$$\hat{\mathbf{F}}_{Z,t} := \hat{\mathbf{Q}}_r' \hat{\mathbf{L}}_t \hat{\mathbf{Q}}_c = \hat{\mathbf{Q}}_r' \mathbf{Y}_t \hat{\mathbf{Q}}_c, \quad \hat{\mathbf{C}}_t := \hat{\mathbf{Q}}_r \hat{\mathbf{F}}_{Z,t} \hat{\mathbf{Q}}_c' = \hat{\mathbf{Q}}_r \hat{\mathbf{Q}}_r' \mathbf{Y}_t \hat{\mathbf{Q}}_c \hat{\mathbf{Q}}_c'. \quad (3.5)$$

Finally, the residuals $\mathbf{E}_t$ is estimated by

$$\hat{\mathbf{E}}_t := \hat{\mathbf{L}}_t - \hat{\mathbf{C}}_t. \quad (3.6)$$

Remark 1 Suppose we have a traditional matrix-valued factor model such that $\dot{\mathbf{Y}}_t = \dot{\mathbf{C}}_t + \dot{\mathbf{E}}_t$ where $\dot{\mathbf{Y}}_t$, $\dot{\mathbf{C}}_t$, and $\dot{\mathbf{E}}_t$ are $p \times q$ matrices representing the observation, common component, and noise, respectively. Suppose also $\dot{\mathbf{C}}_t = \mathbf{A}_r \mathbf{F}_t \mathbf{A}_c'$. Then we can construct

$$\dot{\mu}_t := (pq)^{-1} \mathbf{1}_p' \dot{\mathbf{C}}_t \mathbf{1}_q, \quad \dot{\boldsymbol{\alpha}}_t := q^{-1} \dot{\mathbf{C}}_t \mathbf{1}_q - \dot{\mu}_t \mathbf{1}_p = q^{-1} \mathbf{M}_p \dot{\mathbf{C}}_t \mathbf{1}_q, \quad \dot{\boldsymbol{\beta}}_t' := p^{-1} \dot{\mathbf{C}}_t' \mathbf{1}_p - \dot{\mu}_t \mathbf{1}_q = p^{-1} \mathbf{M}_q \dot{\mathbf{C}}_t' \mathbf{1}_p.$$

Hence we can express FM in the following MEFM form satisfying (IC1):

$$\dot{\mathbf{Y}}_t = \dot{\mu}_t \mathbf{1}_p \mathbf{1}'_q + \dot{\alpha}_t \mathbf{1}'_q + \mathbf{1}_p \dot{\beta}'_t + (\dot{\mathbf{C}}_t - \dot{\mu}_t \mathbf{1}_p \mathbf{1}'_q - \dot{\alpha}_t \mathbf{1}'_q - \mathbf{1}_p \dot{\beta}'_t) + \dot{\mathbf{E}}_t,$$

where

$$\dot{\mathbf{C}}_t - \dot{\mu}_t \mathbf{1}_p \mathbf{1}'_q - \dot{\alpha}_t \mathbf{1}'_q - \mathbf{1}_p \dot{\beta}'_t = (\mathbf{M}_p \mathbf{A}_r) \mathbf{F}_t (\mathbf{M}_q \mathbf{A}_c)',$$

is the common component. Since $\mathbf{M}_m \mathbf{1}_m = \mathbf{0}$, it is easy to see that

$$\mathbf{1}'_p (\mathbf{M}_p \mathbf{A}_r) = \mathbf{0}, \quad \mathbf{1}'_q (\mathbf{M}_q \mathbf{A}_c) = \mathbf{0}.$$

It is also easy to verify that $\mathbf{1}'_p \dot{\alpha}_t = \mathbf{1}'_q \dot{\beta}_t = 0$. Hence a traditional matrix-valued factor model can be expressed as MEFM in (3.1) to satisfy (IC1).

4 Assumptions and Theoretical Results

4.1 Assumptions

A set of assumptions on the factor structure is imposed below, and in particular, we allow factors to have different strengths, as in Lam and Yao (2012) and Chen and Lam (2024).

(M1) (Alpha mixing) *The elements in $\text{vec}(\mathbf{F}_t)$ and $\text{vec}(\mathbf{E}_t)$ are α -mixing. A vector process $\{\mathbf{x}_t : t = 0, \pm 1, \pm 2, \dots\}$ is α -mixing if, for some $\gamma > 2$, the mixing coefficients satisfy the condition that*

$$\sum_{h=1}^{\infty} \alpha(h)^{1-2/\gamma} < \infty,$$

where $\alpha(h) = \sup_{\tau} \sup_{A \in \mathcal{H}_{-\infty}^{\tau}, B \in \mathcal{H}_{\tau+h}^{\infty}} |\mathbb{P}(A \cap B) - \mathbb{P}(A)\mathbb{P}(B)|$ and $\mathcal{H}_{\tau}^s$ is the σ -field generated by $\{\mathbf{x}_t : \tau \leq t \leq s\}$.

(F1) (Time Series in $\mathbf{F}_t$) *There is $\mathbf{X}_{f,t}$ the same dimension as $\mathbf{F}_t$, such that $\mathbf{F}_t = \sum_{w \geq 0} a_{f,w} \mathbf{X}_{f,t-w}$. The time series $\{\mathbf{X}_{f,t}\}$ has i.i.d. elements with mean 0 and variance 1, with uniformly bounded fourth order moments. The coefficients $a_{f,w}$ are such that $\sum_{w \geq 0} a_{f,w}^2 = 1$ and $\sum_{w \geq 0} |a_{f,w}| \leq c$ for some constant c .*

(L1) (Factor strength) *We assume that $\mathbf{A}_r$ and $\mathbf{A}_c$ are of full rank and independent of factors and errors series. Furthermore, as $p, q \rightarrow \infty$,*

$$\mathbf{Z}_r^{-1/2} \mathbf{A}'_r \mathbf{A}_r \mathbf{Z}_r^{-1/2} \rightarrow \Sigma_{A,r}, \quad \mathbf{Z}_c^{-1/2} \mathbf{A}'_c \mathbf{A}_c \mathbf{Z}_c^{-1/2} \rightarrow \Sigma_{A,c}, \quad (4.1)$$

where $\mathbf{Z}_r = \text{diag}(\mathbf{A}'_r \mathbf{A}_r)$, $\mathbf{Z}_c = \text{diag}(\mathbf{A}'_c \mathbf{A}_c)$, and both $\Sigma_{A,r}$ and $\Sigma_{A,c}$ are positive definite with all eigenvalues bounded away from 0 and infinity. We assume $(\mathbf{Z}_r)_{jj} \asymp p^{\delta_{r,j}}$ for $j \in [k_r]$ and $1/2 < \delta_{r,k_r} \leq \dots \leq \delta_{r,2} \leq \delta_{r,1} \leq 1$. Similarly, we assume $(\mathbf{Z}_c)_{jj} \asymp p^{\delta_{c,j}}$ for $j \in [k_c]$, with $1/2 < \delta_{c,k_c} \leq \dots \leq \delta_{c,2} \leq \delta_{c,1} \leq 1$.

With Assumption (L1), we can denote $\mathbf{Q}_r := \mathbf{A}_r \mathbf{Z}_r^{-1/2}$ and $\mathbf{Q}_c := \mathbf{A}_c \mathbf{Z}_c^{-1/2}$. Hence $\mathbf{Q}'_r \mathbf{Q}_r \rightarrow \Sigma_{A,r}$ and $\mathbf{Q}'_c \mathbf{Q}_c \rightarrow \Sigma_{A,c}$.

(E1) (Decomposition of $\mathbf{E}_t$) We assume that

$$\mathbf{E}_t = \mathbf{A}_{e,r} \mathbf{F}_{e,t} \mathbf{A}'_{e,c} + \Sigma_\epsilon \circ \boldsymbol{\epsilon}_t, \quad (4.2)$$

where $\mathbf{F}_{e,t}$ is a matrix of size $k_{e,r} \times k_{e,c}$, containing independent elements with mean 0 and variance 1. The matrix $\boldsymbol{\epsilon}_t \in \mathbb{R}^{p \times q}$ contains independent elements with mean 0 and variance 1, with $\{\boldsymbol{\epsilon}_t\}$ independent of $\{\mathbf{F}_{e,t}\}$. The matrix Σ_ϵ contains the standard deviations of the corresponding elements in $\boldsymbol{\epsilon}_t$, and has elements uniformly bounded away from 0 and infinity.

Moreover, $\mathbf{A}_{e,r}$ and $\mathbf{A}_{e,c}$ are (approximately) sparse matrices with sizes $p \times k_{e,r}$ and $q \times k_{e,c}$ respectively, such that $\|\mathbf{A}_{e,r}\|_1, \|\mathbf{A}_{e,c}\|_1 = O(1)$, with $k_{e,r}, k_{e,c} = O(1)$.

(E2) (Time Series in $\mathbf{E}_t$) There is $\mathbf{X}_{e,t}$ the same dimension as $\mathbf{F}_{e,t}$, and $\mathbf{X}_{\epsilon,t}$ the same dimension as $\boldsymbol{\epsilon}_t$, such that $\mathbf{F}_{e,t} = \sum_{w \geq 0} a_{e,w} \mathbf{X}_{e,t-w}$ and $\boldsymbol{\epsilon}_t = \sum_{w \geq 0} a_{\epsilon,w} \mathbf{X}_{\epsilon,t-w}$, with $\{\mathbf{X}_{e,t}\}$ and $\{\mathbf{X}_{\epsilon,t}\}$ independent of each other, and each time series has independent elements with mean 0 and variance 1 with uniformly bounded fourth order moments. Both $\{\mathbf{X}_{e,t}\}$ and $\{\mathbf{X}_{\epsilon,t}\}$ are independent of $\{\mathbf{X}_{f,t}\}$ from (F1).

The coefficients $a_{e,w}$ and $a_{\epsilon,w}$ are such that $\sum_{w \geq 0} a_{e,w}^2 = \sum_{w \geq 0} a_{\epsilon,w}^2 = 1$ and $\sum_{w \geq 0} |a_{e,w}|, \sum_{w \geq 0} |a_{\epsilon,w}| \leq c$ for some constant c .

(R1) (Rate assumptions) We assume that,

$$\begin{aligned} T^{-1} p^{2(1-\delta_{r,k_r})} q^{1-2\delta_{c,1}} &= o(1), & p^{1-2\delta_{r,k_r}} q^{2(1-\delta_{c,1})} &= o(1), \\ T^{-1} q^{2(1-\delta_{c,k_c})} p^{1-2\delta_{r,1}} &= o(1), & q^{1-2\delta_{c,k_c}} p^{2(1-\delta_{r,1})} &= o(1). \end{aligned}$$

Assumption (F1) introduces serial dependence into the factors, and (E1) and (E2) introduce both cross-sectional and temporal dependence in the noise. The factor structure depicted by (F1), (E1) and (E2) is the same as the one in Cen and Lam (2024). Note that although Assumption (M1) also features in serial dependence, it is mainly used to construct asymptotic normality of estimators.

By Assumption (L1), we have $\mathbf{A}_r \mathbf{F}_t \mathbf{A}_c' = \mathbf{Q}_r \mathbf{Z}_r^{1/2} \mathbf{F}_t \mathbf{Z}_c^{1/2} \mathbf{Q}_c'$, so we aim to estimate $(\mathbf{Q}_r, \mathbf{Q}_c, \mathbf{F}_{Z,t})$ where $\mathbf{F}_{Z,t} := \mathbf{Z}_r^{1/2} \mathbf{F}_t \mathbf{Z}_c^{1/2}$. Unlike the traditional approximate factor models which assumes all factors are pervasive, we allow factors to have different strength similar to Lam and Yao (2012) and Chen and Lam (2024). To be precise, a column of $\mathbf{A}_r$ (resp. $\mathbf{A}_c$) is dense (i.e., a pervasive factor) if the corresponding $\delta_{r,j} = 1$ (resp. $\delta_{c,j} = 1$), otherwise the column represents a weak factor as it is sparse to certain extent.

Due to the presence of potentially weak factors, we require rate conditions in Assumption (R1) for consistency to hold. If all factors are pervasive (R1) holds trivially. We point out that the first and second conditions in (R1) are exactly the same as the first and third conditions of Assumption (R1) in Cen and Lam (2024) for matrix time series.

4.2 Identification of the model

With Assumptions (IC1) and (L1), the model (3.1) is identified according to Theorem 1 below.

Theorem 1 (*Identification*) *With Assumption (IC1), each μ_t , α_t , and β_t can be identified. The common component is hence identified, and if (L1) is also satisfied, the factor structure is identified up to some rotation such that $(\mathbf{Q}_r, \mathbf{Q}_c, \mathbf{F}_{Z,t}) = (\mathbf{Q}_r \mathbf{M}_r, \mathbf{Q}_c \mathbf{M}_c, \mathbf{M}_r^{-1} \mathbf{F}_{Z,t} \mathbf{M}_c^{-1})$ for some invertible matrices $\mathbf{M}_r \in \mathbb{R}^{k_r \times k_r}$ and $\mathbf{M}_c \in \mathbb{R}^{k_c \times k_c}$.*

4.3 Rate of convergence for various estimators

To present the consistency of the loading estimators, define

$$\mathbf{H}_r := T^{-1} \widehat{\mathbf{D}}_r^{-1} \widehat{\mathbf{Q}}_r' \mathbf{Q}_r \sum_{t=1}^T (\mathbf{F}_{Z,t} \mathbf{Q}_c' \mathbf{Q}_c \mathbf{F}_{Z,t}'), \quad (4.3)$$

$$\mathbf{H}_c := T^{-1} \widehat{\mathbf{D}}_c^{-1} \widehat{\mathbf{Q}}_c' \mathbf{Q}_c \sum_{t=1}^T (\mathbf{F}_{Z,t}' \mathbf{Q}_r' \mathbf{Q}_r \mathbf{F}_{Z,t}), \quad (4.4)$$

where $\widehat{\mathbf{D}}_r := \widehat{\mathbf{Q}}_r' (T^{-1} \sum_{t=1}^T \widehat{\mathbf{L}}_t \widehat{\mathbf{L}}_t') \widehat{\mathbf{Q}}_r$ is the $k_r \times k_r$ diagonal matrix of eigenvalues of $T^{-1} \sum_{t=1}^T \widehat{\mathbf{L}}_t \widehat{\mathbf{L}}_t'$, and similarly $\widehat{\mathbf{D}}_c := \widehat{\mathbf{Q}}_c' (T^{-1} \sum_{t=1}^T \widehat{\mathbf{L}}_t \widehat{\mathbf{L}}_t') \widehat{\mathbf{Q}}_c$ is the $k_c \times k_c$ diagonal matrix of eigenvalues of $T^{-1} \sum_{t=1}^T \widehat{\mathbf{L}}_t \widehat{\mathbf{L}}_t'$.

Theorem 2 *Under Assumptions (IC1), (M1), (F1), (L1), (E1), (E2) and (R1), we have*

$$(\hat{\mu}_t - \mu_t)^2 = O_P(p^{-1}q^{-1}), \quad (4.5)$$

$$p^{-1}\|\hat{\alpha}_t - \alpha_t\|^2 = O_P(q^{-1}), \quad (4.6)$$

$$q^{-1}\|\hat{\beta}_t - \beta_t\|^2 = O_P(p^{-1}), \quad (4.7)$$

$$p^{-1}\|\hat{\mathbf{Q}}_r - \mathbf{Q}_r \mathbf{H}_r'\|_F^2 = O_P\left(T^{-1}p^{1-2\delta_{r,k_r}}q^{1-2\delta_{c,1}} + p^{-2\delta_{r,k_r}}q^{2(1-\delta_{c,1})}\right), \quad (4.8)$$

$$q^{-1}\|\hat{\mathbf{Q}}_c - \mathbf{Q}_c \mathbf{H}_c'\|_F^2 = O_P\left(T^{-1}q^{1-2\delta_{c,k_c}}p^{1-2\delta_{r,1}} + q^{-2\delta_{c,k_c}}p^{2(1-\delta_{r,1})}\right). \quad (4.9)$$

From Theorem 2, the consistency for the loading matrix estimators requires Assumption (R1). If all factors are pervasive, the (squared) convergence rates for the row (resp. column) loading matrix will be $\max(1/(Tpq), 1/p^2)$ (resp. $\max(1/(Tpq), 1/q^2)$), which are consistent with those in Chen and Fan (2023) after the same normalization of the loading matrices.

Theorem 3 *Under the assumptions in Theorem 2, we have the following:*

1. *The error of the estimated factor series has rate*

$$\begin{aligned} \|\hat{\mathbf{F}}_{Z,t} - (\mathbf{H}_r^{-1})'\mathbf{F}_{Z,t}\mathbf{H}_c^{-1}\|_F^2 &= O_P(p^{1-\delta_{r,k_r}}q^{1-\delta_{c,k_c}} + T^{-1}p^{1+2\delta_{r,1}-2\delta_{r,k_r}}q^{1-\delta_{c,1}} + p^{1+\delta_{r,1}-3\delta_{r,k_r}}q^{2-\delta_{c,1}} \\ &\quad + T^{-1}q^{1+2\delta_{c,1}-2\delta_{c,k_c}}p^{1-\delta_{r,1}} + q^{1+\delta_{c,1}-3\delta_{c,k_c}}p^{2-\delta_{r,1}}). \end{aligned}$$

2. *For any $t \in [T], i \in [p], j \in [q]$, the squared error of the estimated individual common component is*

$$\begin{aligned} (\hat{C}_{t,ij} - C_{t,ij})^2 &= O_P(p^{1-2\delta_{r,k_r}}q^{1-2\delta_{c,k_c}} + T^{-1}p^{1+2\delta_{r,1}-3\delta_{r,k_r}}q^{1-\delta_{c,1}-\delta_{c,k_c}} + p^{1+\delta_{r,1}-4\delta_{r,k_r}}q^{2-\delta_{c,1}-\delta_{c,k_c}} \\ &\quad + T^{-1}q^{1+2\delta_{c,1}-3\delta_{c,k_c}}p^{1-\delta_{r,1}-\delta_{r,k_r}} + q^{1+\delta_{c,1}-4\delta_{c,k_c}}p^{2-\delta_{r,1}-\delta_{r,k_r}}). \end{aligned}$$

We state the above results separating from Theorem 2 since they have used some arguments from the proof of Theorem 5. If all factors are pervasive, it is clear that individual common components are consistent with rate $(pq)^{-1/2} + T^{-1/2}(q^{-1/2} + p^{-1/2}) + p^{-1} + q^{-1} = \max(1/(Tq)^{1/2}, 1/(Tp)^{1/2}, 1/p, 1/q)$. This rate coincides with Theorem 4 of Chen and Fan (2023) for instance.

4.4 Asymptotic normality of estimators

We present the asymptotic normality of various estimators in this section, together with the estimation of the corresponding covariance matrices for practical inferences. But before that, we need three more assumptions.

(L2) (Eigenvalues) *The eigenvalues of the $k_r \times k_r$ matrix $\Sigma_{A,r} \mathbf{Z}_r$ from Assumption (L1) are distinct, and so are those of the $k_c \times k_c$ matrix $\Sigma_{A,c} \mathbf{Z}_c$.*

(AD1) *There is a constant C such that for any $k \in [K], j \in [d_k]$, as $p, q, T \rightarrow \infty$,*

$$\begin{aligned} \sqrt{\frac{1}{Tqp^{\delta_{r,1}}}} \cdot \mathbb{E} \left\{ \left\| \mathbf{H}_r^* \sum_{i=1}^p \mathbf{Q}_{r,i} \sum_{t=1}^T (\mathbf{C}_t \mathbf{E}'_t)_{ij} \right\| \right\} &\geq C > 0, \\ \sqrt{\frac{1}{Tpq^{\delta_{c,1}}}} \cdot \mathbb{E} \left\{ \left\| \mathbf{H}_c^* \sum_{i=1}^q \mathbf{Q}_{c,i} \sum_{t=1}^T (\mathbf{C}'_t \mathbf{E}_t)_{ij} \right\| \right\} &\geq C > 0, \end{aligned}$$

where $\mathbf{H}_r^* := \text{tr}(\mathbf{A}'_c \mathbf{A}_c)^{1/2} \cdot \mathbf{D}_r^{-1/2} (\mathbf{\Gamma}_r^*)' \mathbf{Z}_r^{1/2}$ with $\mathbf{D}_r := \text{tr}(\mathbf{A}'_c \mathbf{A}_c) \text{diag}\{\lambda_1(\mathbf{A}'_r \mathbf{A}_r), \dots, \lambda_{k_r}(\mathbf{A}'_r \mathbf{A}_r)\}$, and $\mathbf{\Gamma}_r^*$ is the eigenvector matrix of $\text{tr}(\mathbf{A}'_c \mathbf{A}_c) \cdot p^{-\delta_{r,k_r}} q^{-\delta_{c,1}} \mathbf{Z}_r^{1/2} \Sigma_{A,r} \mathbf{Z}_r^{1/2}$. Similarly, we have $\mathbf{H}_c^* := \text{tr}(\mathbf{A}'_r \mathbf{A}_r)^{1/2} \cdot \mathbf{D}_c^{-1/2} (\mathbf{\Gamma}_c^*)' \mathbf{Z}_c^{1/2}$, with $\mathbf{D}_c := \text{tr}(\mathbf{A}'_r \mathbf{A}_r) \text{diag}\{\lambda_1(\mathbf{A}'_c \mathbf{A}_c), \dots, \lambda_{k_c}(\mathbf{A}'_c \mathbf{A}_c)\}$, and $\mathbf{\Gamma}_c^*$ is the eigenvector matrix of $\text{tr}(\mathbf{A}'_r \mathbf{A}_r) \cdot q^{-\delta_{c,k_c}} p^{-\delta_{r,1}} \mathbf{Z}_c^{1/2} \Sigma_{A,c} \mathbf{Z}_c^{1/2}$.

(R2) We have

$$\begin{aligned} T^{-1} p^{1+2\delta_{r,1}-3\delta_{r,k_r}} q^{1-\delta_{c,1}-\delta_{c,k_c}}, \quad p^{1+\delta_{r,1}-4\delta_{r,k_r}} q^{2-\delta_{c,1}-\delta_{c,k_c}}, \\ T^{-1} q^{1+2\delta_{c,1}-3\delta_{c,k_c}} p^{1-\delta_{r,1}-\delta_{r,k_r}}, \quad q^{1+\delta_{c,1}-4\delta_{c,k_c}} p^{2-\delta_{r,1}-\delta_{r,k_r}} = o(1). \end{aligned}$$

Assumption (AD1) appears in Cen and Lam (2024) as well. This assumption facilitates the proof of the asymptotic normality of each row of $\widehat{\mathbf{Q}}_r$ and $\widehat{\mathbf{Q}}_c$, by asserting that in the decomposition of $\widehat{\mathbf{Q}}_r - \mathbf{Q}_r \mathbf{H}_r$ (resp. $\widehat{\mathbf{Q}}_c - \mathbf{Q}_c \mathbf{H}_c$), certain terms are dominating others even in the lower bound, and hence is truly dominating rather than just having the upper bounds dominating other upper bounds as in the proofs of similar theorems in the broader literature of factor models. Assumption (R2) is needed to make sure that the estimated common component $\widehat{\mathbf{C}}_t$ is consistent element-wise (see Theorem 3). This is satisfied automatically when all factors are pervasive, for instance.

Let $\Sigma_{\epsilon,ij}$ be the (i, j) entry of Σ_ϵ in Assumption (E1).

Theorem 4 *Let all assumptions in Theorem 2 hold. Assume also for $i \in [p]$ and $j \in [q]$,*

$$\gamma_{\alpha,i}^2 := \lim_{q \rightarrow \infty} \frac{1}{q} \sum_{j=1}^q \Sigma_{\epsilon,ij}^2, \quad \gamma_{\beta,j}^2 := \lim_{p \rightarrow \infty} \frac{1}{p} \sum_{i=1}^p \Sigma_{\epsilon,ij}^2, \quad \gamma_\mu^2 := \lim_{p,q \rightarrow \infty} \frac{1}{pq} \sum_{i \in [p], j \in [q]} \Sigma_{\epsilon,ij}^2.$$

Then for each $t \in [T]$,

$$\sqrt{pq}(\widehat{\mu}_t - \mu_t) \xrightarrow{\mathcal{D}} \mathcal{N}(0, \gamma_\mu^2).$$

Take a finite integer m , and integers $i_1 < i_2 < \dots < i_m$ with $i_\ell \in [p]$. Define $\boldsymbol{\theta}_{\alpha,t} := (\alpha_{t,i_1}, \dots, \alpha_{t,i_m})'$ and

similarly for $\hat{\boldsymbol{\theta}}_{\alpha,t}$, where $\alpha_{t,i}$ is the i -th element of $\boldsymbol{\alpha}_t$. Then for a fixed $t \in [T]$,

$$\sqrt{q}(\hat{\boldsymbol{\theta}}_{\alpha,t} - \boldsymbol{\theta}_{\alpha,t}) \xrightarrow{\mathcal{D}} \mathcal{N}(\mathbf{0}, \text{diag}(\gamma_{\alpha,i_1}^2, \dots, \gamma_{\alpha,i_m}^2)).$$

Similarly, take integers $j_1 < \dots < j_m$ where $j_\ell \in [q]$. Define $\boldsymbol{\theta}_{\beta,t} := (\beta_{t,j_1}, \dots, \beta_{t,j_m})'$ and similarly for $\hat{\boldsymbol{\theta}}_{\beta,t}$, where $\beta_{t,j}$ is the j -th element of $\boldsymbol{\beta}_t$. Then for a fixed $t \in [T]$,

$$\sqrt{p}(\hat{\boldsymbol{\theta}}_{\beta,t} - \boldsymbol{\theta}_{\beta,t}) \xrightarrow{\mathcal{D}} \mathcal{N}(\mathbf{0}, \text{diag}(\gamma_{\beta,j_1}^2, \dots, \gamma_{\beta,j_m}^2)).$$

Moreover, for $i \in [p]$ and $j \in [q]$, if the rate for $\hat{C}_{t,ij} - C_{t,ij}$ in Theorem 3 is $o(1)$, then

$$\hat{\gamma}_{\alpha,i}^2 := q^{-1}(\hat{\mathbf{E}}_t \hat{\mathbf{E}}_t')_{ii}, \quad \hat{\gamma}_{\beta,j}^2 := p^{-1}(\hat{\mathbf{E}}_t' \hat{\mathbf{E}}_t)_{jj}, \quad \hat{\gamma}_\mu^2 := p^{-1} \sum_{i=1}^p \hat{\gamma}_{\alpha,i}^2 = q^{-1} \sum_{j=1}^q \hat{\gamma}_{\beta,j}^2$$

are consistent estimators for $\gamma_{\alpha,i}^2$, $\gamma_{\beta,j}^2$ and γ_μ^2 respectively under Assumption (R2), so that

$$\begin{aligned} \sqrt{pq} \hat{\gamma}_\mu^{-1}(\hat{\mu}_t - \mu_t) &\xrightarrow{\mathcal{D}} \mathcal{N}(0, 1), \\ \sqrt{q} \text{diag}(\hat{\gamma}_{\alpha,i_1}^{-1}, \dots, \hat{\gamma}_{\alpha,i_m}^{-1})(\hat{\boldsymbol{\theta}}_{\alpha,t} - \boldsymbol{\theta}_{\alpha,t}) &\xrightarrow{\mathcal{D}} \mathcal{N}(\mathbf{0}, \mathbf{I}_m), \\ \sqrt{p} \text{diag}(\hat{\gamma}_{\beta,j_1}^{-1}, \dots, \hat{\gamma}_{\beta,j_m}^{-1})(\hat{\boldsymbol{\theta}}_{\beta,t} - \boldsymbol{\theta}_{\beta,t}) &\xrightarrow{\mathcal{D}} \mathcal{N}(\mathbf{0}, \mathbf{I}_m). \end{aligned}$$

Recall from Remark 1 that FM can be expressed in MEFM, and hence the ability to make inferences on the elements of $\boldsymbol{\alpha}_t$ and $\boldsymbol{\beta}_t$ does not facilitate a test for the necessity of MEFM over FM. For such a test, please see Section 4.5. Theorem 4 gives us the ability to infer on the level of row and column main effects at each time point, which is important if we have target comparisons we want to make for these effects. For instance, if each row represents a country, we can easily compare the main effects at time t for the first country against the average of the second and third simply by considering $\mathbf{g} := (1, -1/2, -1/2)'$, $\boldsymbol{\theta}_{\alpha,t} := (\alpha_{t,1}, \alpha_{t,2}, \alpha_{t,3})'$ and using Theorem 4 to arrive at

$$\sqrt{q}(\mathbf{g}' \text{diag}(\hat{\gamma}_{\alpha,1}^2, \hat{\gamma}_{\alpha,2}^2, \hat{\gamma}_{\alpha,3}^2) \mathbf{g})^{-1/2} \mathbf{g}'(\hat{\boldsymbol{\theta}}_{\alpha,t} - \boldsymbol{\theta}_{\alpha,t}) \xrightarrow{\mathcal{D}} \mathcal{N}(0, 1).$$

Theorem 5 *Let all the assumptions under Theorem 2 hold, in addition to (AD1) and (L2). Suppose k_r and k_c are fixed and $p, q, T \rightarrow \infty$. If $Tq = o(p^{\delta_{r,1} + \delta_{r,k_r}})$, we have*

$$(Tp^{2\delta_{r,k_r} - \delta_{r,1}} q^{2\delta_{c,1} - 1})^{1/2} \cdot (\hat{\mathbf{Q}}_{r,j} - \mathbf{H}_r \mathbf{Q}_{r,j}) \xrightarrow{\mathcal{D}} \mathcal{N}(\mathbf{0}, T^{-1} p^{2\delta_{r,k_r} - \delta_{r,1}} q^{2\delta_{c,1} - 1} \cdot \mathbf{D}_r^{-1} \mathbf{H}_r^* \boldsymbol{\Xi}_{r,j} (\mathbf{H}_r^*)' \mathbf{D}_r^{-1}),$$

$$\text{where } \boldsymbol{\Xi}_{r,j} := \text{plim}_{p,q,T \rightarrow \infty} \text{Var}\left(\sum_{i=1}^p \mathbf{Q}_{r,i} \cdot \sum_{t=1}^T (\mathbf{C}_t \mathbf{E}_t')_{ij}\right).$$

On the other hand, if $Tp = o(q^{\delta_{c,1} + \delta_{c,k_c}})$, we have

$$(Tq^{2\delta_{c,k_c} - \delta_{c,1}} p^{2\delta_{r,1} - 1})^{1/2} \cdot (\widehat{\mathbf{Q}}_{c,j\cdot} - \mathbf{H}_c \mathbf{Q}_{c,j\cdot}) \xrightarrow{\mathcal{D}} \mathcal{N}(\mathbf{0}, T^{-1} q^{2\delta_{c,k_c} - \delta_{c,1}} p^{2\delta_{r,1} - 1} \cdot \mathbf{D}_c^{-1} \mathbf{H}_c^* \boldsymbol{\Xi}_{c,j} (\mathbf{H}_c^*)' \mathbf{D}_c^{-1}),$$

where $\boldsymbol{\Xi}_{c,j} := \text{plim}_{p,q,T \rightarrow \infty} \text{Var}(\sum_{i=1}^q \mathbf{Q}_{c,i\cdot} \sum_{t=1}^T (\mathbf{C}_t' \mathbf{E}_t)_{ij})$.

Theorem 5 is essentially Theorem 3 of Cen and Lam (2024) when $K = 2$ and $\eta = 0$ (no missing values), having the same rate of convergence under potentially weak factors. Hence our MEFM estimation procedure has successfully estimated and removed all time-varying main effects and grand mean, leaving the estimation of the common component exactly the same as in FM.

4.4.1 Estimation of covariance matrix for factor loading estimators

To practically use Theorem 5 for inference, we need to estimate the covariance matrices for $\widehat{\mathbf{Q}}_{r,j\cdot} - \mathbf{H}_r \mathbf{Q}_{r,j\cdot}$ and $\widehat{\mathbf{Q}}_{c,j\cdot} - \mathbf{H}_c \mathbf{Q}_{c,j\cdot}$. We use the heteroscedasticity and autocorrelation consistent (HAC) estimators (Newey and West (1987)) based on $\{\widehat{\mathbf{D}}_r, \widehat{\mathbf{Q}}_r, \widehat{\mathbf{C}}_t, \widehat{\mathbf{E}}_t\}_{t \in [T]}$ and $\{\widehat{\mathbf{D}}_c, \widehat{\mathbf{Q}}_c, \widehat{\mathbf{C}}_t, \widehat{\mathbf{E}}_t\}_{t \in [T]}$, respectively.

For $\widehat{\mathbf{Q}}_{r,j\cdot} - \mathbf{H}_r \mathbf{Q}_{r,j\cdot}$, with η_r such that $\eta_r \rightarrow \infty$ and $\eta_r / (Tp^{2\delta_{r,k_r} - \delta_{r,1}} q^{2\delta_{c,1} - 1})^{1/4} \rightarrow 0$, define an HAC estimator

$$\widehat{\boldsymbol{\Sigma}}_{r,j}^{HAC} := \mathbf{D}_{r,0,j} + \sum_{\nu=1}^{\eta_r} \left(1 - \frac{\nu}{1 + \eta_r}\right) (\mathbf{D}_{r,\nu,j} + \mathbf{D}_{r,\nu,j}'), \quad \text{where}$$

$$\mathbf{D}_{r,\nu,j} := \sum_{t=1+\nu}^T \left\{ \sum_{i=1}^p \left(T^{-1} \widehat{\mathbf{D}}_r^{-1} \widehat{\mathbf{Q}}_r' \sum_{s=1}^T \widehat{\mathbf{C}}_s \widehat{\mathbf{C}}_{s,i} \right) (\widehat{\mathbf{C}}_t \widehat{\mathbf{E}}_t)_{ij} \right\} \left\{ \sum_{i=1}^p \left(T^{-1} \widehat{\mathbf{D}}_r^{-1} \widehat{\mathbf{Q}}_r' \sum_{s=1}^T \widehat{\mathbf{C}}_s \widehat{\mathbf{C}}_{s,i} \right) (\widehat{\mathbf{C}}_{t-\nu} \widehat{\mathbf{E}}_{t-\nu})_{ij} \right\}'.$$

Similarly for $\widehat{\mathbf{Q}}_{c,j\cdot} - \mathbf{H}_c \mathbf{Q}_{c,j\cdot}$, with η_c such that $\eta_c \rightarrow \infty$ and $\eta_c / (Tq^{2\delta_{c,k_c} - \delta_{c,1}} p^{2\delta_{r,1} - 1})^{1/4} \rightarrow 0$, define

$$\widehat{\boldsymbol{\Sigma}}_{c,j}^{HAC} := \mathbf{D}_{c,0,j} + \sum_{\nu=1}^{\eta_c} \left(1 - \frac{\nu}{1 + \eta_c}\right) (\mathbf{D}_{c,\nu,j} + \mathbf{D}_{c,\nu,j}'), \quad \text{where}$$

$$\mathbf{D}_{c,\nu,j} := \sum_{t=1+\nu}^T \left\{ \sum_{i=1}^q \left(T^{-1} \widehat{\mathbf{D}}_c^{-1} \widehat{\mathbf{Q}}_c' \sum_{s=1}^T \widehat{\mathbf{C}}_s \widehat{\mathbf{C}}_{s,i} \right) (\widehat{\mathbf{C}}_t \widehat{\mathbf{E}}_t)_{ij} \right\} \left\{ \sum_{i=1}^q \left(T^{-1} \widehat{\mathbf{D}}_c^{-1} \widehat{\mathbf{Q}}_c' \sum_{s=1}^T \widehat{\mathbf{C}}_s \widehat{\mathbf{C}}_{s,i} \right) (\widehat{\mathbf{C}}_{t-\nu} \widehat{\mathbf{E}}_{t-\nu})_{ij} \right\}'.$$

Theorem 6 *Let all the assumptions under Theorem 2 hold, in addition to (L2), (AD1) and (R2). Suppose k_r and k_c are fixed and $p, q, T \rightarrow \infty$. If $Tq = o(p^{\delta_{r,1} + \delta_{r,k_r}})$, then*

1. $\widehat{\mathbf{D}}_r^{-1} \widehat{\boldsymbol{\Sigma}}_{r,j}^{HAC} \widehat{\mathbf{D}}_r^{-1}$ is consistent for $\mathbf{D}_r^{-1} \mathbf{H}_r^* \boldsymbol{\Xi}_{r,j} (\mathbf{H}_r^*)' \mathbf{D}_r^{-1}$;
2. $T \cdot (\widehat{\boldsymbol{\Sigma}}_{r,j}^{HAC})^{-1/2} \widehat{\mathbf{D}}_r (\widehat{\mathbf{Q}}_{r,j\cdot} - \mathbf{H}_r \mathbf{Q}_{r,j\cdot}) \xrightarrow{\mathcal{D}} \mathcal{N}(\mathbf{0}, \mathbf{I}_{k_r})$.

On the other hand, if $Tp = o(q^{\delta_{c,1} + \delta_{c,k_c}})$, then

3. $\widehat{\mathbf{D}}_c^{-1} \widehat{\boldsymbol{\Sigma}}_{c,j}^{HAC} \widehat{\mathbf{D}}_c^{-1}$ is consistent for $\mathbf{D}_c^{-1} \mathbf{H}_c^* \boldsymbol{\Xi}_{c,j} (\mathbf{H}_c^*)' \mathbf{D}_c^{-1}$;

$$4. \quad T \cdot (\widehat{\Sigma}_{c,j}^{HAC})^{-1/2} \widehat{\mathbf{D}}_c (\widehat{\mathbf{Q}}_{c,j} - \mathbf{H}_c \mathbf{Q}_{c,j}) \xrightarrow{\mathcal{D}} \mathcal{N}(\mathbf{0}, \mathbf{I}_{k_c}).$$

4.5 Testing the sufficiency of FM versus MEFM

In the last section, we introduce how to make inferences on various parameters of MEFM. However, to test if FM is sufficient against our proposed MEFM, simple inferences on the model parameters are not enough in the face of Remark 1. Formally, we want to test, for the time horizon $t \in [T]$,

$$H_0 : \text{FM is sufficient over } t \in [T] \quad \longleftrightarrow \quad H_1 : \text{MEFM is needed over } t \in [T].$$

The above problem is complicated by the fact that, in Section 3.1, we have seen that MEFM can always be expressed as FM if we are willing to potentially consider a large number of factors. So, how “large” an increase in the number of factors do we consider unacceptable?

Remark 1 tells us that a special form of MEFM can be expressed back in FM:

$$\mathbf{Y}_t = \mu_t \mathbf{1}_p \mathbf{1}_q' + \boldsymbol{\alpha}_t \mathbf{1}_q' + \mathbf{1}_p \boldsymbol{\beta}_t' + \mathbf{M}_p \dot{\mathbf{C}}_t \mathbf{M}_q + \mathbf{E}_t = \mathbf{A}_r \mathbf{F}_t \mathbf{A}_c' + \mathbf{E}_t, \quad t \in [T],$$

where $\dot{\mathbf{C}}_t := \mathbf{A}_r \mathbf{F}_t \mathbf{A}_c'$ and

$$\mu_t := (pq)^{-1} \mathbf{1}_p' \dot{\mathbf{C}}_t \mathbf{1}_q, \quad \boldsymbol{\alpha}_t := q^{-1} \mathbf{M}_p \dot{\mathbf{C}}_t \mathbf{1}_q, \quad \boldsymbol{\beta}_t := p^{-1} \mathbf{M}_q \dot{\mathbf{C}}_t' \mathbf{1}_p.$$

If $\mathbf{A}_r$ has rank k_r satisfying Assumption (L1) and $\mathbf{A}_c$ has rank k_c , the potential rank of $\mathbf{M}_p \mathbf{A}_r$ is $k_r - 1$ (when a column in $\mathbf{A}_r$ is parallel to $\mathbf{1}_p$), and that of $\mathbf{M}_q \mathbf{A}_c$ is $k_c - 1$ (when a column in $\mathbf{A}_c$ is parallel to $\mathbf{1}_q$), demonstrating that FM can have an increase in the number of factors, albeit still finite.

Another special example is when both $\boldsymbol{\alpha}_t$ and $\boldsymbol{\beta}_t$ are zero, but $\mu_t \neq 0$. Then we can write MEFM as

$$\mathbf{Y}_t = \mu_t \mathbf{1}_p \mathbf{1}_q' + \mathbf{A}_r \mathbf{F}_t \mathbf{A}_c' + \mathbf{E}_t = (\mathbf{A}_r, \mathbf{1}_p) \begin{pmatrix} \mathbf{F}_t & \mathbf{0} \\ \mathbf{0}' & \mu_t/(pq) \end{pmatrix} \begin{pmatrix} \mathbf{A}_c' \\ \mathbf{1}_q' \end{pmatrix} + \mathbf{E}_t,$$

which is FM with loading matrices $(\mathbf{A}_r, \mathbf{1}_p)$ and $(\mathbf{A}_c, \mathbf{1}_q)$ respectively, and an increase by 1 for both the number of row and column factors.

In light of the above examples, we deem FM sufficient if and only if the number of factors in the FM is still finite and any model variables satisfy the Assumptions in Section 4.1.

To be able to test H_0 against H_1 , define $\tilde{\mathbf{E}}_t$ to be the residual matrix after a fitting of FM (a similar

procedure to fitting MEFM but treating μ_t , α_t and β_t as zero), with

$$\check{\mathbf{E}}_t := \mathbf{Y}_t - \check{\mathbf{C}}_t, \quad \text{where } \check{\mathbf{C}}_t := \check{\mathbf{A}}_r \check{\mathbf{A}}_r' \mathbf{Y}_t \check{\mathbf{A}}_c \check{\mathbf{A}}_c',$$

with $\check{\mathbf{A}}_r$ and $\check{\mathbf{A}}_c$ the $p \times \ell_r$ and $q \times \ell_c$ eigenmatrices of $\sum_{t=1}^T \mathbf{Y}_t \mathbf{Y}_t'$ and $\sum_{t=1}^T \mathbf{Y}_t' \mathbf{Y}_t$ respectively.

Theorem 7 *Let all assumptions in Theorem 2 hold, on top of (R2). Also assume that $\widehat{C}_{t,ij} - C_{t,ij} = o_P(\min(p^{-1/2}, q^{-1/2}))$ in Theorem 3. Suppose k_r, k_c, ℓ_r and ℓ_c are all fixed and known. Then under H_0 , for each $t \in [T]$, we have*

$$\begin{aligned} \frac{(\widehat{\mathbf{E}}_t \widehat{\mathbf{E}}_t')_{ii} - \sum_{j=1}^q \Sigma_{\epsilon, ij}^2}{\sqrt{\sum_{j=1}^q \text{Var}(\epsilon_{t, ij}^2) \Sigma_{\epsilon, ij}^4}}, \quad \frac{(\check{\mathbf{E}}_t \check{\mathbf{E}}_t')_{ii} - \sum_{j=1}^q \Sigma_{\epsilon, ij}^2}{\sqrt{\sum_{j=1}^q \text{Var}(\epsilon_{t, ij}^2) \Sigma_{\epsilon, ij}^4}} &\xrightarrow{\mathcal{D}} Z_i \xrightarrow{\mathcal{D}} N(0, 1) \text{ for each } i \in [p]; \\ \frac{(\widehat{\mathbf{E}}_t \widehat{\mathbf{E}}_t)_{jj} - \sum_{i=1}^p \Sigma_{\epsilon, ij}^2}{\sqrt{\sum_{i=1}^p \text{Var}(\epsilon_{t, ij}^2) \Sigma_{\epsilon, ij}^4}}, \quad \frac{(\check{\mathbf{E}}_t \check{\mathbf{E}}_t)_{jj} - \sum_{i=1}^p \Sigma_{\epsilon, ij}^2}{\sqrt{\sum_{i=1}^p \text{Var}(\epsilon_{t, ij}^2) \Sigma_{\epsilon, ij}^4}} &\xrightarrow{\mathcal{D}} W_j \xrightarrow{\mathcal{D}} N(0, 1) \text{ for each } j \in [q], \end{aligned}$$

where Z_h is independent of Z_ℓ and W_h is independent of W_ℓ for $h \neq \ell$. The same asymptotic results hold true under H_1 for $(\widehat{\mathbf{E}}_t \widehat{\mathbf{E}}_t')_{ii}$ and $(\widehat{\mathbf{E}}_t' \widehat{\mathbf{E}}_t)_{jj}$ respectively for $i \in [p], j \in [q]$.

The assumption $\widehat{C}_{t,ij} - C_{t,ij} = o_P(\min(p^{-1/2}, q^{-1/2}))$ is satisfied, for instance, when all factors are pervasive and T, p, q are of the same order. Theorem 7 tells us that for each $t \in [T]$, both

$$x_{\alpha, t} := \max_{i \in [p]} \widehat{\gamma}_{\alpha, i}^2 = \max_{i \in [p]} \{q^{-1} (\widehat{\mathbf{E}}_t \widehat{\mathbf{E}}_t')_{ii}\}, \quad y_{\alpha, t} := \max_{i \in [p]} \check{\gamma}_{\alpha, i}^2 := \max_{i \in [p]} \{q^{-1} (\check{\mathbf{E}}_t \check{\mathbf{E}}_t')_{ii}\} \quad (4.10)$$

are distributed approximately the same for large q under H_0 , and $x_{\alpha, t}$ in particular is distributed the same no matter under H_0 or H_1 . Similarly, define

$$x_{\beta, t} := \max_{j \in [q]} \widehat{\gamma}_{\beta, j}^2 = \max_{j \in [q]} \{p^{-1} (\widehat{\mathbf{E}}_t' \widehat{\mathbf{E}}_t)_{jj}\}, \quad y_{\beta, t} := \max_{j \in [q]} \check{\gamma}_{\beta, j}^2 := \max_{j \in [q]} \{p^{-1} (\check{\mathbf{E}}_t' \check{\mathbf{E}}_t)_{jj}\}, \quad (4.11)$$

which are distributed approximately the same for large p under H_0 from Theorem 7, and $x_{\beta, t}$ in particular is distributed the same no matter under H_0 or H_1 .

Define $\mathbb{F}_{x, \alpha}$, $\mathbb{F}_{y, \alpha}$, $\mathbb{F}_{x, \beta}$ and $\mathbb{F}_{y, \beta}$ the empirical cumulative distribution functions for $\{x_{\alpha, t}\}_{t \in [T]}$, $\{y_{\alpha, t}\}_{t \in [T]}$, $\{x_{\beta, t}\}_{t \in [T]}$ and $\{y_{\beta, t}\}_{t \in [T]}$ respectively:

$$\begin{aligned} \mathbb{F}_{x, \alpha}(c) &:= \frac{1}{T} \sum_{t=1}^T \mathbf{1}\{x_{\alpha, t} \leq c\}, \quad \mathbb{F}_{y, \alpha}(c) := \frac{1}{T} \sum_{t=1}^T \mathbf{1}\{y_{\alpha, t} \leq c\}, \\ \mathbb{F}_{x, \beta}(c) &:= \frac{1}{T} \sum_{t=1}^T \mathbf{1}\{x_{\beta, t} \leq c\}, \quad \mathbb{F}_{y, \beta}(c) := \frac{1}{T} \sum_{t=1}^T \mathbf{1}\{y_{\beta, t} \leq c\}. \end{aligned} \quad (4.12)$$

Corollary 8 *Let all the assumptions in Theorem 7 hold. Define for $0 < \theta < 1$,*

$$\hat{q}_{x,\alpha}(\theta) := \inf\{c \mid \mathbb{F}_{x,\alpha}(c) \geq \theta\}, \quad \hat{q}_{x,\beta}(\theta) := \inf\{c \mid \mathbb{F}_{x,\beta}(c) \geq \theta\},$$

Then under H_0 , as $T, p, q \rightarrow \infty$, we have for each $t \in [T]$,

$$\mathbb{P}_{y,\alpha}[y_{\alpha,t} \geq \hat{q}_{x,\alpha}(\theta)] \leq 1 - \theta, \quad \mathbb{P}_{y,\beta}[y_{\beta,t} \geq \hat{q}_{x,\beta}(\theta)] \leq 1 - \theta,$$

where $\mathbb{P}_{y,\alpha}$ and $\mathbb{P}_{y,\beta}$ are empirical probability measures induced by $\mathbb{F}_{y,\alpha}$ and $\mathbb{F}_{y,\beta}$ respectively.

With Corollary 8, we can test H_0 at significance level $1 - \theta$ using the test statistics $y_{\alpha,t}$ and $y_{\beta,t}$, and rejection rules $y_{\alpha,t} \geq \hat{q}_{x,\alpha}(\theta)$ and $y_{\beta,t} \geq \hat{q}_{x,\beta}(\theta)$ respectively. Since we have $y_{\alpha,t}$ and $y_{\beta,t}$ for $t \in [T]$, we can assess the significance level under H_0 by calculating

$$\text{Significance levels} = T^{-1} \sum_{t=1}^T \mathbf{1}\{y_{\alpha,t} \geq \hat{q}_{x,\alpha}(\theta)\}, \quad T^{-1} \sum_{t=1}^T \mathbf{1}\{y_{\beta,t} \geq \hat{q}_{x,\beta}(\theta)\},$$

and see if they are close to $1 - \theta$. If H_0 is not true, then if any $\alpha_{t,i}$ is large, we expect $y_{\alpha,t}$ to be large. Or, if any $\beta_{t,j}$ is large, we expect $y_{\beta,t}$ to be large.

In practice for testing H_0 against H_1 , we estimate k_r and k_c , and set $\ell_r = k_r + 1$ and $\ell_c = k_c + 1$ in light of the previous argument on how a special form of MEFM can be expressed back in FM. For estimation of k_r and k_c , see Section 4.6.

4.6 Estimation of the number of factors

From (3.4), we have $T^{-1} \sum_{t=1}^T \hat{\mathbf{L}}_t \hat{\mathbf{L}}_t'$ essentially the row sample covariance matrix and $T^{-1} \sum_{t=1}^T \hat{\mathbf{L}}_t' \hat{\mathbf{L}}_t$ the column sample covariance matrix. We then propose the eigenvalue-ratio estimators for the number of factors as

$$\hat{k}_r := \arg \min_j \left\{ \frac{\lambda_{j+1}(T^{-1} \sum_{t=1}^T \hat{\mathbf{L}}_t \hat{\mathbf{L}}_t') + \xi_r}{\lambda_j(T^{-1} \sum_{t=1}^T \hat{\mathbf{L}}_t \hat{\mathbf{L}}_t') + \xi_r}, j \in \llbracket p/2 \rrbracket \right\}, \quad \xi_r \asymp pq[(Tq)^{-1/2} + p^{-1/2}], \quad (4.13)$$

$$\hat{k}_c := \arg \min_j \left\{ \frac{\lambda_{j+1}(T^{-1} \sum_{t=1}^T \hat{\mathbf{L}}_t' \hat{\mathbf{L}}_t) + \xi_c}{\lambda_j(T^{-1} \sum_{t=1}^T \hat{\mathbf{L}}_t' \hat{\mathbf{L}}_t) + \xi_c}, j \in \llbracket q/2 \rrbracket \right\}, \quad \xi_c \asymp pq[(Tp)^{-1/2} + q^{-1/2}]. \quad (4.14)$$

Ratio-based estimators are widely studied by researchers. For example, an eigenvalue-ratio estimator is considered in Lam and Yao (2012) and Ahn and Horenstein (2013), while a cumulative eigenvalue ratio estimator is proposed by Zhang et al. (2023). Our proposed estimator is similar to the perturbed eigenvalue-ratio estimators as in Pelger (2019). Technically, we can minimise (4.13) (resp. (4.14) over any $j \in [p]$ (resp. $j \in [q]$), but it is very reasonable to assume $k_r \leq p/2$ and $k_c \leq q/2$ in all applications of factor

models. The correction terms ξ_r and ξ_c are added to stabilise the ratio so that consistency follows from the theorem below.

Theorem 9 *Under Assumptions (IC1), (M1), (F1), (L1), (E1), (E2) and (R1), we have the following.*

1. $\hat{k}_r$ is a consistent estimator of k_r if

$$\begin{cases} p^{1-\delta_{r,k_r}} q^{1-\delta_{c,1}} [(Tq)^{-1/2} + p^{-1/2}] = o(p^{\delta_{r,j+1}-\delta_{r,j}}), & j \in [k_r - 1] \text{ with } k_r \geq 2; \\ p^{1-\delta_{r,1}} q^{1-\delta_{c,1}} [(Tq)^{-1/2} + p^{-1/2}] = o(1), & k_r = 1. \end{cases}$$

2. $\hat{k}_c$ is a consistent estimator of k_c if

$$\begin{cases} q^{1-\delta_{c,k_c}} p^{1-\delta_{r,1}} [(Tp)^{-1/2} + q^{-1/2}] = o(q^{\delta_{c,j+1}-\delta_{c,j}}), & j \in [k_c - 1] \text{ with } k_c \geq 2; \\ q^{1-\delta_{c,1}} p^{1-\delta_{r,1}} [(Tp)^{-1/2} + q^{-1/2}] = o(1), & k_c = 1. \end{cases}$$

The extra rate conditions in the theorem are due to existence of potential weak factors and are trivially satisfied for pervasive factors. The theorem is similar to the consistency result in Cen and Lam (2024) for matrix-valued factor models, and this implies that the number of factors in MEFM can be well estimated just as in the case of FM.

5 Numerical Results

5.1 Simulation

We demonstrate the performance of our estimators in this section. We will experiment different settings to assess consistency results as described in Theorem 2 and 3, followed by the asymptotic normality of our estimators in Theorem 4 and 5, where the covariance matrices can be constructed by their consistent estimators by Theorem 4 and Theorem 6, respectively. We then showcase the results for the rank estimators described in Theorem 9. As it is a first to consider matrix factor model with time-varying grand mean and main effects, we unveil the differences between MEFM and FM using numerical results that will illustrate Theorem 7.

For the data generating process, we use Assumptions (E1), (E2) and (F1) to generate general linear processes for the noise and factor series in model (3.1). To be precise, the elements in $\mathbf{F}_t$ are independent standardised AR(5) with AR coefficients 0.7, 0.3, -0.4, 0.2, and -0.1. The elements in $\mathbf{F}_{e,t}$ and $\boldsymbol{\epsilon}_t$ are generated similarly, but their AR coefficients are (-0.7, -0.3, -0.4, 0.2, 0.1) and (0.8, 0.4, -0.4, 0.2, -0.1) respectively. The standard deviation of each element in $\boldsymbol{\epsilon}_t$ is generated by i.i.d. $|\mathcal{N}(0,1)|$. To test how

robust our method is under heavy-tailed distribution, we consider two distributions for the innovation process in generating $\mathbf{F}_t$, $\mathbf{F}_{e,t}$ and $\boldsymbol{\epsilon}_t$: 1) i.i.d. $\mathcal{N}(0, 1)$; 2) i.i.d. t_3 .

The row factor loading matrix $\mathbf{A}_r$ is generated with $\mathbf{A}_r = \mathbf{M}_p \mathbf{U}_r \mathbf{B}_r$, where each entry of $\mathbf{U}_r \in \mathbb{R}^{p \times k_r}$ is i.i.d. $\mathcal{N}(0, 1)$, and $\mathbf{B}_r \in \mathbb{R}^{k_r \times k_r}$ is diagonal with the j -th diagonal entry being $p^{-\zeta_{r,j}}$, $0 \leq \zeta_{r,j} \leq 0.5$. Pervasive (strong) factors have $\zeta_{r,j} = 0$, while weak factors have $0 < \zeta_{r,j} \leq 0.5$. Note that $\mathbf{M}_p$ is defined in (3.4) so that (IC1) is satisfied. In a similar way, the column factor loading matrix $\mathbf{A}_c$ is generated independently. Each entry of $\mathbf{A}_{e,r} \in \mathbb{R}^{p \times k_{e,r}}$ is i.i.d. $\mathcal{N}(0, 1)$ and has independent probability of 0.95 being set exactly to 0, and $\mathbf{A}_{e,c}$ is generated similarly. We fix $k_{e,r} = k_{e,c} = 2$ throughout the section.

For any $t \in [T]$, we generate $\mu_t = v_{\mu,t}$, $\boldsymbol{\alpha}_t = \mathbf{M}_p \mathbf{v}_{\alpha,t}$ and $\boldsymbol{\beta}_t = \mathbf{M}_q \mathbf{v}_{\beta,t}$, where $v_{\mu,t}$ is $\mathcal{N}(m_\mu, \sigma_\mu^2)$, each element of $\mathbf{v}_{\alpha,t}$ is i.i.d. $\mathcal{N}(m_\alpha, \sigma_\alpha^2)$ and that of $\mathbf{v}_{\beta,t}$ is i.i.d. $\mathcal{N}(m_\beta, \sigma_\beta^2)$. We set $m_\mu = m_\alpha = m_\beta = 0$ and $\sigma_\mu = \sigma_\alpha = \sigma_\beta = 1$, and every experiment in this section is repeated 1000 times unless specified otherwise.

5.1.1 Accuracy of various estimators

To assess the accuracy of our estimators, we define the relative mean squared errors (MSE) for μ_t , $\boldsymbol{\alpha}_t$, $\boldsymbol{\beta}_t$ and $\mathbf{C}_t$ as the following, respectively,

$$\begin{aligned} \text{relative MSE}_\mu &= \frac{\sum_{t=1}^T (\mu_t - \hat{\mu}_t)^2}{\sum_{t=1}^T \mu_t^2}, & \text{relative MSE}_\alpha &= \frac{\sum_{t=1}^T \|\boldsymbol{\alpha}_t - \hat{\boldsymbol{\alpha}}_t\|^2}{\sum_{t=1}^T \|\boldsymbol{\alpha}_t\|^2}, \\ \text{relative MSE}_\beta &= \frac{\sum_{t=1}^T \|\boldsymbol{\beta}_t - \hat{\boldsymbol{\beta}}_t\|^2}{\sum_{t=1}^T \|\boldsymbol{\beta}_t\|^2}, & \text{relative MSE}_\mathbf{C} &= \frac{\sum_{t=1}^T \|\mathbf{C}_t - \hat{\mathbf{C}}_t\|_F^2}{\sum_{t=1}^T \|\mathbf{C}_t\|_F^2}. \end{aligned}$$

For measuring the accuracy of our factor loading matrix estimators, we use the column space distance,

$$\mathcal{D}(\mathbf{Q}, \widehat{\mathbf{Q}}) = \left\| \mathbf{Q}(\mathbf{Q}'\mathbf{Q})^{-1}\mathbf{Q}' - \widehat{\mathbf{Q}}(\widehat{\mathbf{Q}}'\widehat{\mathbf{Q}})^{-1}\widehat{\mathbf{Q}}' \right\|,$$

for any given $\mathbf{Q}$ and $\widehat{\mathbf{Q}}$, which is a common measure in the literature such as Chen et al. (2022) and Chen and Fan (2023).

We consider the following settings:

- (Ia) $T = 100$, $p = q = 40$, $k_r = 1$, $k_c = 2$. All factors are pervasive with $\zeta_{r,j} = \zeta_{c,j} = 0$. All innovation processes in constructing $\mathbf{F}_t$, $\mathbf{F}_{e,t}$ and $\boldsymbol{\epsilon}_t$ are i.i.d. standard normal.
- (Ib) Same as (Ia), but one factor is weak with $\zeta_{r,1} = 0.2$ and $\zeta_{c,1} = 0.2$. Set also $m_\alpha = -2$.
- (Ic) Same as (Ia), but all innovation processes are i.i.d. t_3 .
- (Id) Same as (Ib), but $T = 100$, $p = q = 80$ and $\sigma_\alpha = 2$.
- (Ie) Same as (Id), but $T = 200$.

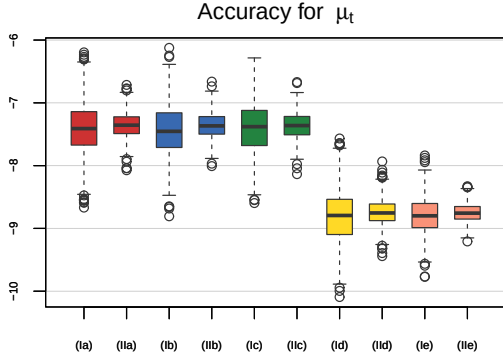


Figure 1: Plot of the relative MSE for μ_t (in log-scale) from Settings (Ia) to (Ie), comparing with (IIa) to (IIe).

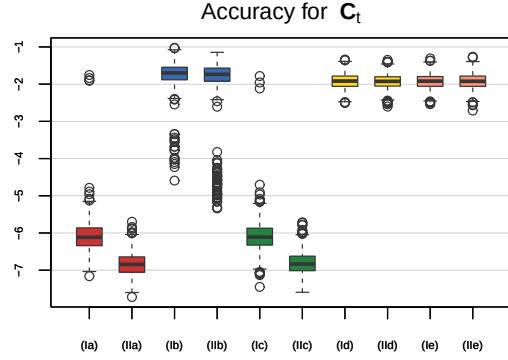


Figure 2: Plot of the relative MSE for $\mathbf{C}_t$ (in log-scale) from Settings (Ia) to (Ie), comparing with (IIa) to (IIe).

(IIa-e) Same as (Ia) to (Ie) respectively, except that we generate $\mathbf{F}_t$, $\mathbf{F}_{e,t}$ and ϵ_t using white noise rather than AR(5).

Setting (IIa) to (IIe) are to investigate how temporal dependence in the noise affects our results.

We report the boxplots of accuracy measures for our estimators from Figure 1 to Figure 6. Note first that stronger temporal dependence leads to larger variance of our estimators in general. The serial dependence mainly undermines the performance of our loading matrix estimators as shown in Figures 5 and 6, which in turn affects our common component estimator.

Considering the comparisons among (Ia) to (Ie), we see that relative MSE_μ can be improved by increasing the spatial dimensions, but is not affected by weak factors. Similar results can be seen from Figure 3 and Figure 4 for relative MSE_α and relative MSE_β . The detrimental effects of heavy-tailed innovation processes in Setting (Ic) are most reflected in the corresponding boxplots in Figure 4.

Weak factors can be detrimental to the accuracy of the factor loading matrix estimators, as can be seen by the significant rise in the factor loading space errors from Setting (Ia) to (Ib) in Figure 5 and 6. In fact, $\hat{k}_c$ barely captures the second factor under Setting (Ib) and (IIb). See Section 5.1.2 for details. Comparing Setting (Ib) with (Id), Figure 5 and 6 show that increase in data dimensions slightly improves our factor loading matrix estimators, which is consistent to the simulation results in Wang et al. (2019) for instance.

5.1.2 Performance for the estimation of the number of factors

In this section, we demonstrate the performance of our estimators for the number of factors, as described in Theorem 9. First, we set $\xi_r = pq[(Tq)^{-1/2} + p^{-1/2}]/5$ and $\xi_c = pq[(Tp)^{-1/2} + q^{-1/2}]/5$, so that the conditions for ξ_r and ξ_c in (4.13) and (4.14) are respectively satisfied. A wide range of values other than 1/5 for ξ_r and ξ_c are experimented, but 1/5 is working the best in vast majority of settings, and hence we

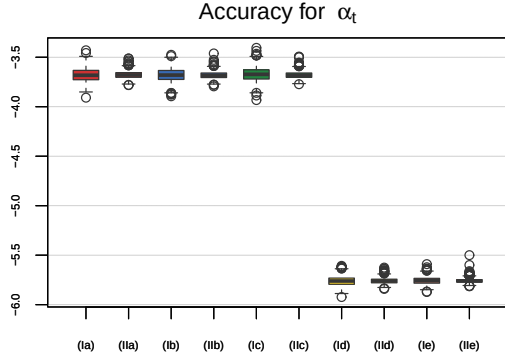


Figure 3: Plot of the relative MSE for α_t (in log-scale) from Settings (Ia) to (Ie), comparing with (IIa) to (IIe).

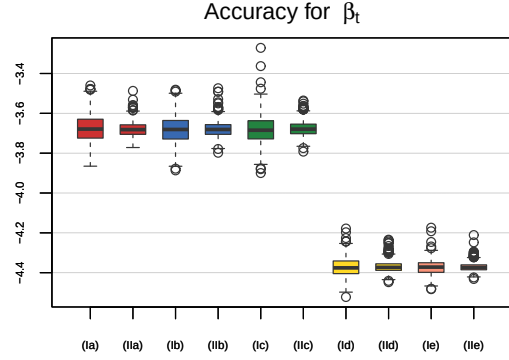


Figure 4: Plot of the relative MSE for β_t (in log-scale) from Settings (Ia) to (Ie), comparing with (IIa) to (IIe).

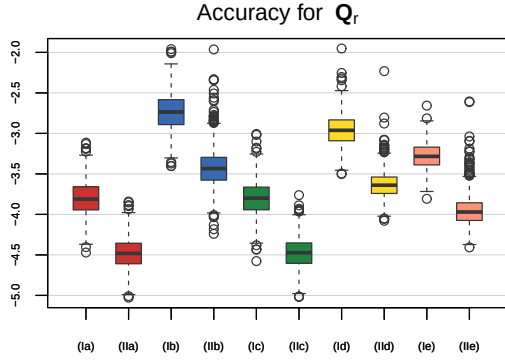


Figure 5: Plot of the row space distance $\mathcal{D}(\mathbf{Q}_r, \widehat{\mathbf{Q}}_r)$ (in log-scale) from Settings (Ia) to (Ie), comparing with (IIa) to (IIe).

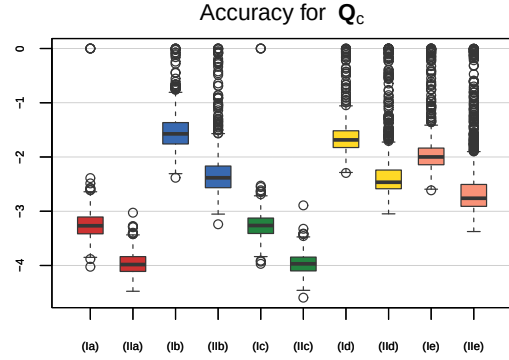


Figure 6: Plot of the column space distance $\mathcal{D}(\mathbf{Q}_c, \widehat{\mathbf{Q}}_c)$ (in log-scale) from Settings (Ia) to (Ie), comparing with (IIa) to (IIe).

do not recommend treating this as a tuning parameter.

We present the results for each of the following settings:

- (IIIa) $k_r = k_c = 3$. All factors are pervasive with $\zeta_{r,j} = \zeta_{c,j} = 0$ for all $j \in [3]$. All innovation processes involved are i.i.d. standard normal.
- (IIIb) Same as (IIIa), but some factors are weak with $\zeta_{r,1} = \zeta_{c,1} = \zeta_{c,2} = 0.2$.
- (IIIc) Same as (IIIa), but all factors are weak with $\zeta_{r,j} = \zeta_{c,j} = 0.2$ for all $j \in [3]$.

We experiment the above settings with various (p, q) pairs among $(10, 10)$, $(10, 20)$ and $(20, 20)$, with the choice $T = 0.5 \cdot pq$ or $T = pq$. The setup is similar to Wang et al. (2019) and Chen and Fan (2023), but we use smaller sets of dimensions since the accuracy of our estimators are approaching 1 with larger dimensions, which reveal little intricacies among different settings.

From the results in Table 1, our eigenvalue-ratio estimators is working well with MEFM. The accuracy of $\widehat{k}_r$ and $\widehat{k}_c$ suffers from the existence of weak factors, which is also seen in traditional FM (see for

$(\hat{k}_r, \hat{k}_c)$	$p, q = 10, 10$		$p, q = 10, 20$		$p, q = 20, 20$	
	$T = .5pq$	$T = pq$	$T = .5pq$	$T = pq$	$T = .5pq$	$T = pq$
	Setting (IIIa)					
(2, 3)	0.121	0.112	0.128	0.11	0	0.004
(3, 2)	0.124	0.111	0.004	0.003	0.001	0.001
(3, 3)	0.583	0.659	0.833	0.855	0.999	0.995
other	0.172	0.118	0.035	0.032	0	0
	Setting (IIIb)					
(2, 3)	0.135	0.13	0.23	0.257	0.228	0.149
(3, 2)	0.079	0.096	0.024	0.017	0.022	0.02
(3, 3)	0.136	0.17	0.289	0.347	0.556	0.637
other	0.65	0.604	0.457	0.379	0.194	0.194
	Setting (IIIc)					
(2, 3)	0.082	0.085	0.218	0.254	0.089	0.096
(3, 2)	0.075	0.124	0.04	0.035	0.088	0.089
(3, 3)	0.073	0.096	0.209	0.257	0.614	0.646
other	0.77	0.695	0.533	0.454	0.209	0.169

Table 1: Results for Setting (IIIa) to (IIIc). Each cell reports the frequency of $(\hat{k}_r, \hat{k}_c)$ under the setting in the corresponding column. The true number of factors is $(k_r, k_c) = (3, 3)$, and the cells corresponding to correct estimations are bolded.

instance Chen and Lam (2024) and Cen and Lam (2024)). In particular, the accuracy of our estimators drops significantly as we move from Setting (IIIa) to (IIIc), and in general large dimensions are beneficial to our estimation. Lastly, note that although we have two weak factors in the column loading matrix while there is only one weak factor in the row loading matrix, the correct proportion of $\hat{k}_c$ is much larger than that of $\hat{k}_r$ for $(p, q) = (10, 20)$. This hints at the importance of data dimensions over factor strength, which can also be seen from the fact that the results for $(p, q) = (20, 20)$ under Setting (IIIc) are comparable with those for $(p, q) = (10, 10)$ under Setting (IIIa).

5.1.3 Asymptotic normality

We numerically demonstrate the asymptotic normality results in Theorems 4 and 5 in this section. For the ease of demonstration, we consider $t = 10$ only for the asymptotic distribution of $\hat{\mu}_t, \hat{\theta}_{\alpha,t} = (\hat{\alpha}_{t,1}, \hat{\alpha}_{t,2}, \hat{\alpha}_{t,3})'$ and $\hat{\theta}_{\beta,t} = (\hat{\beta}_{t,1}, \hat{\beta}_{t,2}, \hat{\beta}_{t,3})'$, and for $\hat{\theta}_{\alpha,t}$ and $\hat{\theta}_{\beta,t}$ we will only report results for the third component. We will also demonstrate the asymptotic normality for $(\hat{\mathbf{Q}}_c)_1$. and present the results for $(\hat{\mathbf{Q}}_c)_{11}$, i.e., the first entry of the first row in the column loading matrix estimator. To consistently estimate its covariance matrix, we use Theorem 6 with $\eta_c = \lfloor (Tpq)^{1/4}/5 \rfloor$.

We use heavy-tailed innovations to investigate the robustness of our results, hence Setting (Ic) is adapted except that we generate $\mathbf{F}_t$, $\mathbf{F}_{c,t}$ and ϵ_t using AR(1) with coefficient -0.2 . Due to the different

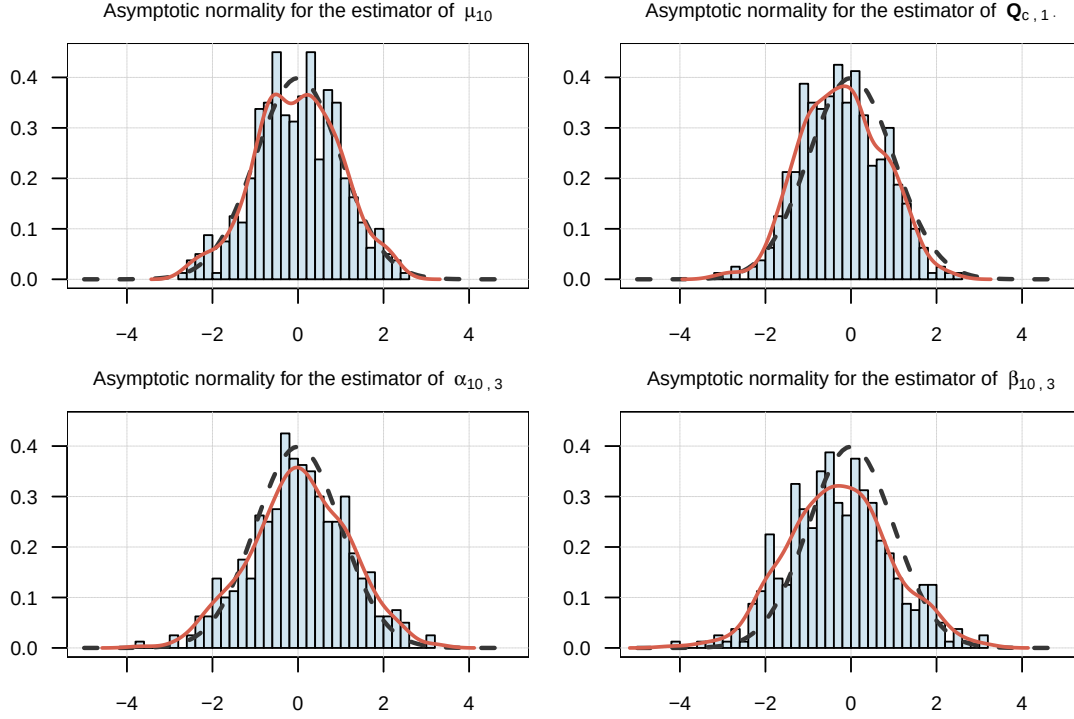


Figure 7: Histograms of $\sqrt{pq}\hat{\gamma}_\mu^{-1}(\hat{\mu}_{10} - \mu_{10})$ (top-left), $[T(\hat{\Sigma}_{c,1}^{HAC})^{-1/2}\hat{\mathbf{D}}_c(\hat{\mathbf{Q}}_{c,1} - \mathbf{H}_1^a\mathbf{Q}_{c,1})]_1$ (top-right), $\sqrt{q}[\text{diag}(\hat{\gamma}_{\alpha,1}^{-1}, \hat{\gamma}_{\alpha,2}^{-1}, \hat{\gamma}_{\alpha,3}^{-1})(\hat{\boldsymbol{\theta}}_{\alpha,10} - \boldsymbol{\theta}_{\alpha,10})]_3$ (bottom-left), and $\sqrt{p}[\text{diag}(\hat{\gamma}_{\beta,1}^{-1}, \hat{\gamma}_{\beta,2}^{-1}, \hat{\gamma}_{\beta,3}^{-1})(\hat{\boldsymbol{\theta}}_{\beta,10} - \boldsymbol{\theta}_{\beta,10})]_3$ (bottom-right). In each panel, the curve (in red) is the empirical density, and the density curve for $\mathcal{N}(0, 1)$ (in black, dotted) is also superimposed.

rates of convergence in Theorems 4 and 5, we specify different dimensions (T, p, q) in the following settings:

$$\hat{\mu}_t : (80, 100, 100), \quad \hat{\boldsymbol{\theta}}_{\alpha,t} : (60, 60, 300), \quad \hat{\boldsymbol{\theta}}_{\beta,t} : (60, 300, 60), \quad (\hat{\mathbf{Q}}_c)_1 : (60, 60, 300),$$

where the dimension setting for $(\hat{\mathbf{Q}}_c)_1$ is to align with the rate conditions in Theorem 5 that $Tp/q^2 \rightarrow 0$ under pervasive factors. Each setting is repeated 400 times, and we present the histograms of our four estimators in Figure 7.

Our plots stand as empirical evidence of Theorem 4, 5 and 6. It might worth noting that the spread of the normalised empirical density for $\hat{\beta}_{10,3}$ is slightly larger than expected by comparing with the superimposed standard normal. The same problem is not seen in the histogram for $\hat{\alpha}_{10,3}$. With true $(k_r, k_c) = (1, 2)$, the common component estimation using $(p, q) = (300, 60)$ is worse than that using $(p, q) = (60, 300)$ due to insufficient column dimension relative to k_c . Hence it leads to worse estimators for errors and $(\hat{\gamma}_{\beta,1}^{-1}, \hat{\gamma}_{\beta,2}^{-1}, \hat{\gamma}_{\beta,3}^{-1})$ under $(p, q) = (300, 60)$. Hence inference performances on the time-varying row and column effect estimators are affected by the latent number of factors.

5.1.4 Testing MEFM versus FM

We demonstrate numerical results for Corollary 8 in this section. We consider the two scenarios below:

1. (*Global effect.*) The entries of at least one of α_t and β_t are in general non-zero for each t .
2. (*Local effect.*) The entries of at least one of α_t and β_t are sparse for each t , i.e., given any t , there are some non-zero entries in at least one of α_t and β_t with all other entries zero.

Throughout this section, we generate the time-varying grand mean and main effects using Rademacher random variables such that $v_{\mu,t}$ is i.i.d. Rademacher multiplied by some u_μ and each entry of $\mathbf{v}_{\alpha,t}$, $\mathbf{v}_{\beta,t}$ is i.i.d. Rademacher multiplied by some u_α , u_β respectively, recalling that $\mu_t = v_{\mu,t}$, $\alpha_t = \mathbf{M}_p \mathbf{v}_{\alpha,t}$ and $\beta_t = \mathbf{M}_q \mathbf{v}_{\beta,t}$. Hence, setting $u_\mu = u_\alpha = u_\beta = 0$ corresponds to generating a traditional FM. We set $k_r = k_c = 2$, and consider the following settings:

- (IVa) $T = p = q = 40$. All factors are pervasive with $\zeta_{r,j} = \zeta_{c,j} = 0$. All innovation processes in constructing $\mathbf{F}_t$, $\mathbf{F}_{e,t}$ and ϵ_t are i.i.d. standard normal. Set $u_\mu = u_\beta = 0$, and we select u_α from 0.1, 0.5, 1.
- (IVb) Same as (IVa), but fix $u_\alpha = 0.1$ and select u_β from 0.1, 0.5, 1.
- (IVc) Same as (IVa), except that $u_\alpha = 1$, and when generating $\alpha_t = \mathbf{M}_p \mathbf{v}_{\alpha,t}$ as specified previously, we only keep the first u_{local} entries of $\mathbf{v}_{\alpha,t}$ as non-zero where u_{local} is selected from 2, 5, 10.

Setting (IVa) and (IVb) are designed for testing global effects, and Setting (IVc) for local effects. For each setting, we construct $y_{\alpha,t}$, $y_{\beta,t}$ and use $\theta = 0.95$ in Corollary 8. Each experiment is repeated 400 times and we report both $\mathbf{reject}_\alpha := T^{-1} \sum_{t=1}^T \mathbb{1}\{y_{\alpha,t} \geq \hat{q}_{x,\alpha}(0.95)\}$ and $\mathbf{reject}_\beta := T^{-1} \sum_{t=1}^T \mathbb{1}\{y_{\beta,t} \geq \hat{q}_{x,\beta}(0.95)\}$.

As explained under Corollary 8, we expect $\mathbf{reject}_\alpha$ and $\mathbf{reject}_\beta$ to be close to $1 - \theta = 0.05$ if FM is sufficient. From Table 2, our proposed test works well since it suggests FM is insufficient as we strengthen α_t or β_t . In particular, even if the signal of α_t is not strong enough such as $u_\alpha = 0.1$, Setting (IVb) shows that additional signals from β_t allows us to reject the use of FM. The comparison between $\mathbf{reject}_\alpha$ and $\mathbf{reject}_\beta$ is indicative of which effect is stronger. According to the results for (IVc) in the table, our test is capable of detecting local effect such that $\mathbf{reject}_\alpha$ is far from 0.05 even when only two entries in α_t are non-zero.

Extensive experiments on different dimensions, factor strengths or grand mean magnitudes are performed. All indicate similar interpretation as the above settings and hence the results are not shown here. The power curve for Setting (IVa) is also presented in Figure 8 to support the use of our test, with $(T, p, q) = (60, 80, 80)$ and u_α ranging from 0.02 to 1. Besides, we also show the power curve for local

Parameter	Size	Setting (IVa)			Setting (IVb)			Setting (IVc)		
	0	0.1	0.5	1	0.1	0.5	1	2	5	10
reject $_{\alpha}$	5 ₍₄₎	11 ₍₇₎	63 ₍₃₁₎	96 ₍₁₅₎	13 ₍₈₎	53 ₍₃₀₎	86 ₍₂₃₎	37 ₍₁₇₎	77 ₍₂₄₎	85 ₍₂₇₎
reject $_{\beta}$	5 ₍₄₎	11 ₍₇₎	52 ₍₂₈₎	87 ₍₂₂₎	13 ₍₈₎	62 ₍₃₂₎	96 ₍₁₆₎	14 ₍₈₎	28 ₍₁₆₎	48 ₍₂₆₎

Table 2: Results for Setting (IVa) to (IVd). Each cell reports the mean and SD (subscripted, in bracket), both multiplied by 100. The parameters for Settings (IVa), (IVb) and (IVc) are u_{α} , u_{β} and u_{local} , respectively. Setting (IVa) with $u_{\alpha} = 0$ is reported in the first column, representing the size of the test.

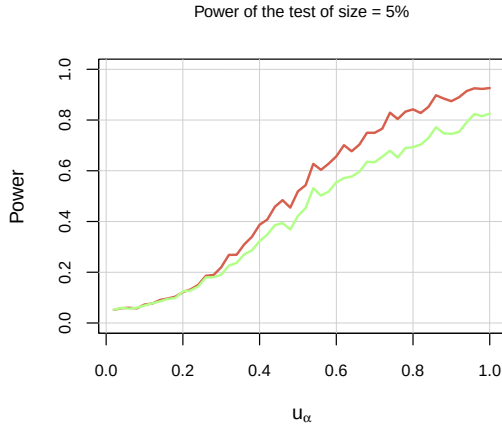


Figure 8: Statistical power curve of testing the null hypothesis that FM is sufficient for the given series, against the alternative that MEFM is necessary. Each power value is computed as the average over 400 runs of **reject** $_{\alpha}$ (in red) and **reject** $_{\beta}$ (in green) under Setting (IVa) except that $(T, p, q) = (60, 80, 80)$.

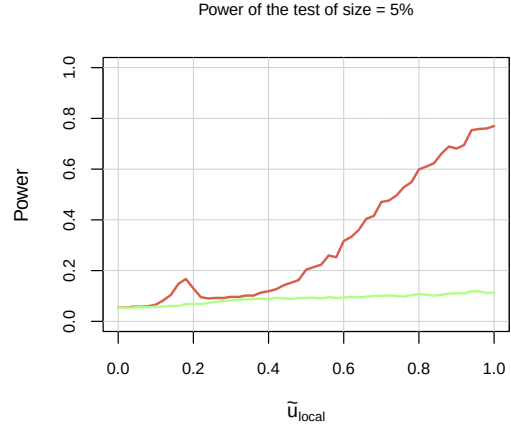


Figure 9: Statistical power curve of testing the null hypothesis that FM is sufficient for the given series, against the alternative that MEFM is necessary. Refer to Figure 8 for how the power is computed. The data is generated under Setting (IVc) except that $(T, p, q) = (60, 80, 80)$ and α_t is generated as $\alpha_1 = \tilde{u}_{local}(1, 1, -2, 0, \dots, 0)'$, $\alpha_2 = \tilde{u}_{local}(1, 2, -3, 0, \dots, 0)'$, $\alpha_3 = \tilde{u}_{local}(2, -5, 3, 0, \dots, 0)'$ and $\alpha_{3\ell+i} = \alpha_i$ for ℓ a positive integer and $i = 1, 2, 3$, so that each α_t has non-zero entries only in the first three indices.

effect in Figure 9, for Setting (IVc) except that $(T, p, q) = (60, 80, 80)$ and we generate α_t as described in the caption. Both power curves show that the test is able to reject the use of FM if signals from the time-varying main effects are large, either globally or locally. And in both figures, when u_{α} is close to 0.02 or $\tilde{u}_{local}$ close to 0, the value of the power curves are all very close to 0.05, which is exactly what we want for the size of the tests.

5.2 Real data analysis

5.2.1 NYC taxi traffic

We analyse a set of taxi traffic data in New York city in this example. The data includes all individual taxi rides operated by Yellow Taxi in New York City, published at

<https://www1.nyc.gov/site/tlc/about/tlc-trip-record-data.page>.

For simplicity, we only consider the rides within Manhattan Island, which comprises most of the data. The dataset contains 842 million trip records within the period of January 1, 2013 to December 31, 2022. Each trip record includes features such as pick-up and drop-off dates/times, pick-up and drop-off locations, trip distances, itemized fares, rate types, payment types, and driver-reported passenger counts. Our example here focuses on the drop-off dates/times and locations.

To classify the drop-off locations in Manhattan, they are coded according to 69 predefined zones in the dataset. Moreover, each day is divided into 24 hourly periods to represent the drop-off times each day, with the first hourly period from 0 a.m. to 1 a.m. The total number of rides moving among the zones within each hour are recorded, yielding data $\mathbf{Y}_t \in \mathbb{R}^{69 \times 24}$ each day, where $y_{i_1, i_2, t}$ is the number of trips to zone i_1 and the pick-up time is within the i_2 -th hourly period on day t .

We consider the non-business-day series which is 1,133 days long, within the period of January 1, 2013 to December 31, 2022. Using MEFM, the estimated rank of the core factors is (2, 2) according to our proposed eigenvalue ratio estimator. As mentioned in Section 4.5, we therefore use (3, 3) as the number of factors to estimate FM and test if FM is sufficient. We compute $\mathbf{reject}_\alpha = 0.064$ and $\mathbf{reject}_\beta = 0.133$ which are defined in Section 5.1.4. They should be close to $1 - \theta = 0.05$ according to Corollary 8 if FM is sufficient. Hence we reject the use of traditional FM due to the signals in $\hat{\beta}_t$.

To compare MEFM with FM, we use core rank (3, 3) to estimate MEFM for the rest of this section. Figure 10 and 11 illustrate the heatmaps of the estimated loading columns on the three dropoff factors using MEFM and FM, respectively. From both heatmaps, we can identify the first factor as active areas, the second as dining and sports areas and the third as downtown areas. The three factors are similar to their corresponding counterparts, except that the first factor estimated using MEFM is more indicative on the active areas to taxi traffic in Manhattan by its emphasised orange zone which corresponds to East Harlem.

To gain further understanding on the taxi traffic, we show the scaled $\hat{\mathbf{Q}}_2$ by MEFM and FM in Tables 3 and 4, respectively. We can see that for the rush hours between 6 p.m. to 11 p.m., the estimated loadings almost vanish for MEFM, which is consistent with the fact that $\hat{\beta}_t$ captures the common hour effect on Manhattan life style. This also provides an intuition why the time-varying column/hour effect is strong, since in non-business days, the way that daily hours affecting the taxi traffic can change drastically over



Figure 10: Estimated loading on three dropoff factors using MEFM, i.e. $\widehat{\mathbf{Q}}_{1,1}$ (left), $\widehat{\mathbf{Q}}_{1,2}$ (middle) and $\widehat{\mathbf{Q}}_{1,3}$ (right).



Figure 11: Estimated loading on three dropoff factors using FM, similar to Figure 10.

time as compared to the same when Manhattan zones are considered. For demonstration purpose, we plot both $\widehat{\beta}_{t,2}$ and $\widehat{\beta}_{t,18}$ in Figure 12, where the former series features the mid-night effects and the latter features the night-life effects. Both series demonstrate obvious seasonality before COVID-19 as indicated on the plot.

The business-day series is also analysed, but since both $\mathbf{reject}_\alpha$ and $\mathbf{reject}_\beta$ are not significant, the estimated model is not shown here. The fact that the time-varying hour effect is not strong for business days is probably due to a rather routine working hours. Thus the hour effect is hardly changing and can

be absorbed into a fixed mean, so that FM would be sufficient.

0 _{am}																																				
	2			4			6			8			10			12 _{pm}			2			4			6			8			10			12 _{am}		
1	-2	-5	-6	-7	-7	-7	-6	-5	-3	0	3	5	6	6	5	5	4	4	5	5	2	0	0	-1												
2	6	5	3	1	-1	-4	-5	-6	-7	-7	-6	-5	-3	-2	-1	-2	-1	-1	2	5	8	6	6	7												
3	-1	-13	-9	-6	-2	2	4	5	6	4	2	-2	-4	-5	-4	-3	-2	-1	0	2	5	4	7	9												

Table 3: Estimated loading matrix $\widehat{\mathbf{Q}}_2$ using MEFM, after scaling. Magnitudes larger than 6 are highlighted in red.

0 _{am}	12 _{pm}																								12 _{am}
	2	4	6	8	10	12 _{pm}	2	4	6	8	10	12 _{am}													
1	-5	-5	-4	-3	-2	-1	-1	-1	-2	-3	-4	-5	-5	-6	-5	-5	-5	-6	-6	-6	-5	-5	-5		
2	5	7	7	5	4	2	0	-2	-4	-6	-6	-6	-6	-5	-4	-4	-3	-3	-2	1	4	4	5	6	
3	1	-13	-10	-9	-6	-3	-1	0	1	0	-1	-3	-3	-3	-3	-2	-1	0	3	6	6	6	8	11	

Table 4: Estimated loading matrix $\widehat{\mathbf{Q}}_2$ using FM, after scaling. Magnitudes larger than 5 are highlighted in red.

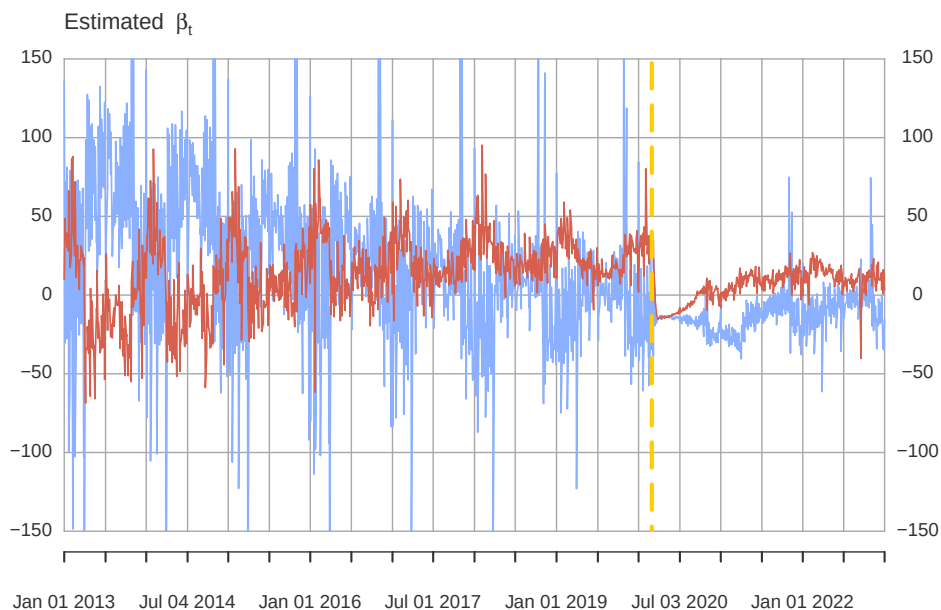


Figure 12: Plot of the estimated hour effects for periods from 1 a.m. to 2 a.m. (in blue) and from 5 p.m. to 6 p.m. (in red). The date for the first confirmed case of COVID-19 in New York is also shown (dotted yellow vertical line).

6 Appendix

Proofs of all the theorems in this paper can be found in the supplement of this paper at <http://stats.lse.ac.uk/lam/Supp-MEFM.pdf>. Instruction in using our R package **MEFM** can be found here.

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